SOLUM Newton Pro

Electronic Shelf Label Datasheet

20/03/2024

Summary

This datasheet presents the general performance and specifications of NEWTON PRO-Label for SOLUM Electronic Shelf Label (ESL) System.

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Document History

Rev.	Date	Revision History	Page
Preliminary	12/14/2023	Preliminary release	-
1.0	03/14/2024	Specification modification	
1.1	03/20/2024	Specification & Label modification	

1 Preface

1.1 About This Guide

This datasheet presents the specification and general performance of SOLUM's Newton PRO Electronic Shelf Labels (ESLs) lineup.

1.2 Audience

This manual is intended for any user (IT, operations, store managers, installers, etc.) authorized to operate and install SOLUM ESLs.

1.3 Abbreviations and Acronyms

Terminology/Abbreviation	Description
GW	Gateway
ESL	Electronic Shelf Label
RF	Radio Frequency
IT	Information Technology
PoE	Power over Ethernet
TBD	To Be Decided

2 Overview

SOLUM Newton PRO Electronic Shelf Labels (ESLs) are components to a total SOLUM's ESL System. The SOLUM's ESL System consists of the ESLs, Gateway(s), Server, and optional accessories (such as the Newton PRO Remote Controller) and is used to electronically displays key information such as price and product information, that are traditionally printed or written on paper in environments like supermarkets, warehouses, and factories.

SOLUM's Newton PRO ESLs are the industry leading solutions that provide the longest battery life (up to 10 years), fastest update speed, built in LEDs, built in button IP68 rating for various environments, multiple pages per ESL, and more to take the operation beyond just displaying information on the ESLs.

Newton PRO ESLs come in various sizes and colors to meet all customer use cases. They are battery powered for ease of deployment and wirelessly receive updates from SOLUM Gateway for.

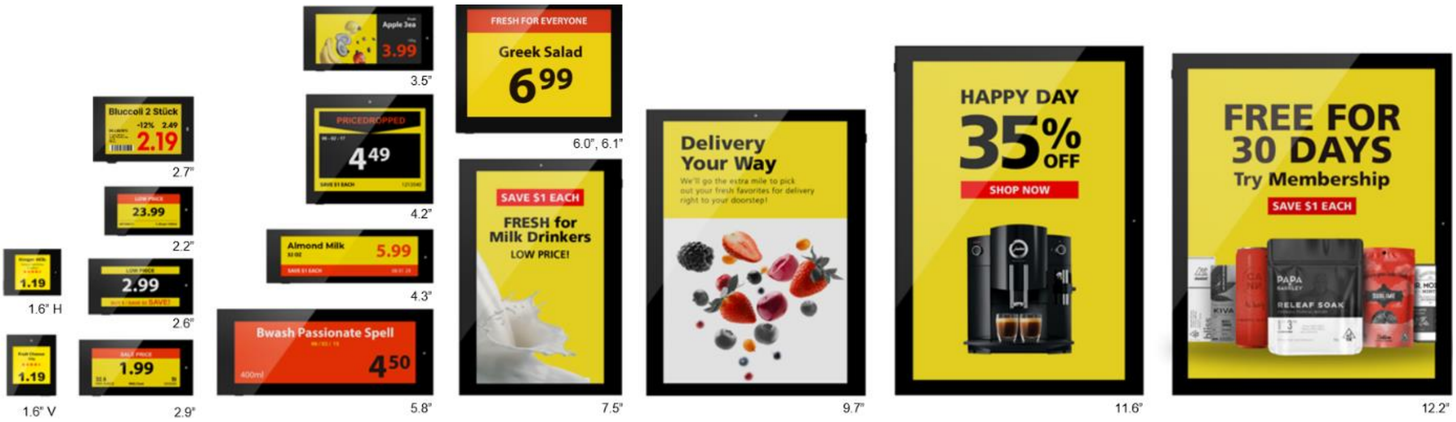


Figure 1. NEWTON PRO PRODUCT LINE-UP

3 Specification

This section details specification of each ESL by size. ESLs are identified by the diagonal measurement of the display in inches. For example, a 2.9" ESL is referring to an ESL with the diagonal display dimension of 2.9".

3.1 Product Specification

Item	Description
Label Dimensions (W x D x H(mm))	**1.6"V:** 36.7 x 44.9 x 10.45 mm / 1.45 x 1.77 x 0.41 inch **1.6"H:** 44.9 x 36.7 x 10.45 mm / 1.77 x 1.45 x 0.41 inch **2.2":** 67.0 x 37.4 x 10.45 mm / 2.64 x 1.47 x 0.41 inch **2.6":** 79.2 x 42.5 x 10.45 mm / 3.12 x 1.67 x 0.41 inch **2.7":** 76.0 x 49.5 x 10.5 mm / 2.99 x 1.95 x 0.41 inch **2.9":** 85.9 x 41.3 x 10.45 mm / 3.38 x 1.62 x 0.41 inch **3.5":** 100.2 x 49.6 x 10.45 mm / 3.94 x 1.95 x 0.41 inch **4.2":** 98.1 x 83.7 x 12.45 mm / 3.86 x 3.29 x 0.49 inch **4.3":** 130.2 x 42.3 x 13.0 mm / 5.13 x 1.67 x 0.51 inch **5.8":** 164.0 x 66.8 x 14.3 mm / 6.45 x 2.63 x 0.56 inch **6.0":** 133.0 x 109.0 x 14.30 mm / 5.23 x 4.29 x 0.56 inch **6.1":** 133.2 × 109.96 × 14.9 mm / 5.24 x 4.33 x 0.58 inch **7.5":** 125.0 x 182.9 x 14.85 mm / 4.92 x 7.2 x 0.58 inch **9.7":** 170.2 x 223.6 x 14.85 mm / 6.7 x 8.8 x 0.58 inch **11.6":** 192.0 x 269.1 x 15.35 mm / 7.50 x 10.6 x 0.60 inch **12.2":** 216.2 x 260.0 x 15.35 mm / 8.51 x 10.23 x 0.60 inch
Display Dimensions (W x D(mm))	**1.6"V:** 200 x 200 Pixel (188dpi) / 27.0 x 27.0mm / 1.06 x 1.06 inch **1.6"H:** 200 x 200 Pixel (188dpi) / 27.0 x 27.0mm / 1.06 x 1.06 inch **2.2":** 296 x 160 Pixel (156dpi) / 48.1 x 26.0mm / 1.89 x 1.02 inch **2.6":** 360 x 184 Pixel (152dpi) / 60.0 x 30.7mm / 2.37 x 1.21 inch **2.7":** 300 x 200 Pixel (133dpi) / 57.3 x 38.2mm / 2.25 x 1.50 inch **2.9":** 384 x 168 Pixel (144dpi) / 67.6 x 29.6 mm / 2.66 x 1.16 inch **3.5":** 384 x 180 Pixel (120dpi) / 81.0 x 38.0 mm / 3.19 x 1.50 inch **4.2":** 400 x 300 Pixel (120dpi) / 84.8 x 63.6 mm / 3.34 x 2.50 inch **4.3":** 522 x 152 Pixel (125dpi) / 105.4 x 30.7 mm / 4.15 x 1.21 inch **5.8":** 792 x 272 Pixel(144dpi) / 139.0 x 47.7 mm / 5.47 x 1.88 inch **6.0":** 600 × 448 Pixel(132dpi) / 114.9 × 85.8 mm / 4.52 × 3.38 inch **6.1":** 648 × 480 Pixel(138dpi) / 118.8 × 88.2 mm / 4.68 × 3.47 inch **7.5":** 480 x 800 Pixel (126dpi) / 97.9 x 163.2 mm / 3.8 x 6.4 inch **9.7":** 672 x 960 Pixel(121dpi) /141.1 x 201.6 mm / 5.56 x 7.94 inch **11.6":** 640 x 960 Pixel (100dpi) / 163.0 x 244.5 mm / 6.42 x 9.63 inch **12.2":** 768 x 960 Pixel(102dpi) / 190.1 x 237.6 mm / 7.48 x 9.35 inch
Label Weight	**1.6"V:** 19.20gr **1.6"H:** 19.75gr **2.2":** 32.65gr **2.6":** 38.40gr **2.7":** 41.70gr **2.9":** 40.25gr **3.5":** 50.00gr **4.2":** 89.90gr **4.3":** 70.80gr **5.8":** 131.00gr **6.0":** 167.00gr **6.1":** 167.00gr **7.5":** 238.60gr **9.7":** 386.60gr

	11.6”: 557.00gr **12.2”:** 586.00gr
Viewing Angle	Nearly 180°
Display Colors	●○●● BWRY (black, white, red, yellow) ** color options are not available for all sizes..
Battery	Type and quantity of batteries defer based on ESL **CR2450 Lithium Battery** 1pcs : 1.6” 2pcs : 2.2”, 2.6”, 2.7”, 2.9”, 3.5”, 4.2” **CR2450 Battery Pack** 3pcs : 4.3” 4pcs : 5.8”, 6.0”, 6.1”, 7.5” 5pcs : 9.7”, 11.6” 6pcs : 12.2”
Wireless Communication	2.4 GHz Unlicensed ISM band BLE physical layer with SOLUM proprietary protocol
Communication Distance	98 feet (30m) radius Line of Sight
Security	128-bit AES Encryption
Compliance	FCC, IC, CE, TELEC, KC, RoHS
Operating Temperature	**Nominal ESLs** **BWRY :** 32°F ~ 104°F (0°C ~ 40°C) @45~70% RH
Storage Temperature	32°F ~ 104°F (0°C ~ 40°C) @45~70% RH

3.2 Radio (RF) Specification

Item	Parameter	Specification			Unit	Condition
		Min	Typ	Max		
Tx	Tx Power		4		dBm	
	[Carrier Frequency Offset and Drift]	-75	0	75	kHz	
	Tx Current		-	10	mA	total current at max Tx Power
Rx	Receiver Sensitivity	-85	-	-	dBm	PER < 5%

3.3 NFC Specification

Item	Parameter	Specification			Unit	Condition
		Min	Typ	Max		
NFC	Read Distance	-	0.7	-	in	
		-	20	-	mm	

NFC antenna location shown for each ESL size.

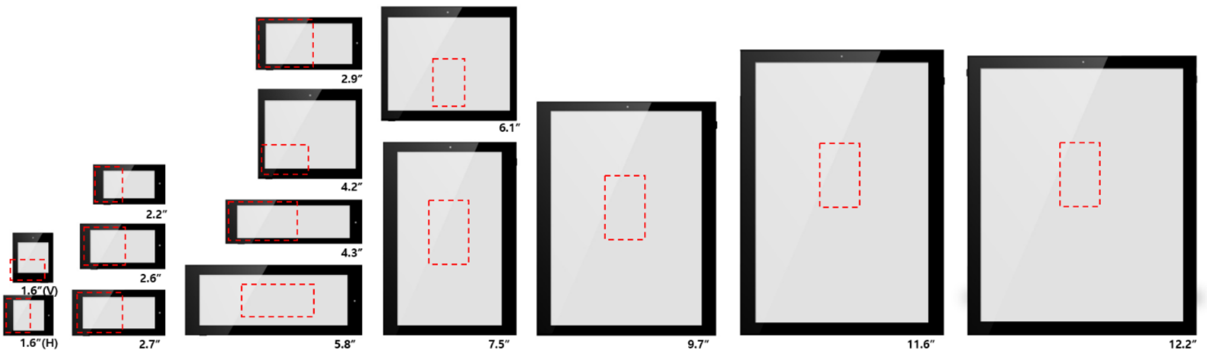


Figure 2. NFC LOCATION

3.4 Features

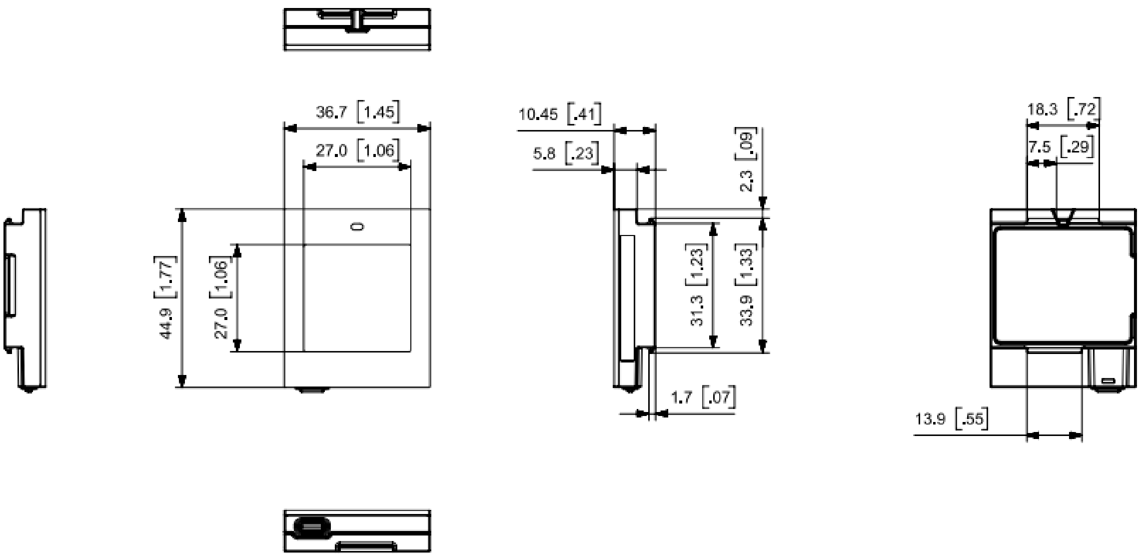
Some of the features of SOLUM ESL.

- Low power consumption (up to 10year battery life)
- ‘Real time’ update speed
- SOLUM proprietary protocol communication with SOLUM Gateway for added security

Item	Description
LED	7 colors (Red, Green, Blue, Yellow, Cyan, Magenta, White)
Usable Pages	7 pages
NFC	Built-in Felica NFC Forum Type 3
IP Rating	IP68
Housing Bezel Color	White, Black

3.5 Mechanical Drawing

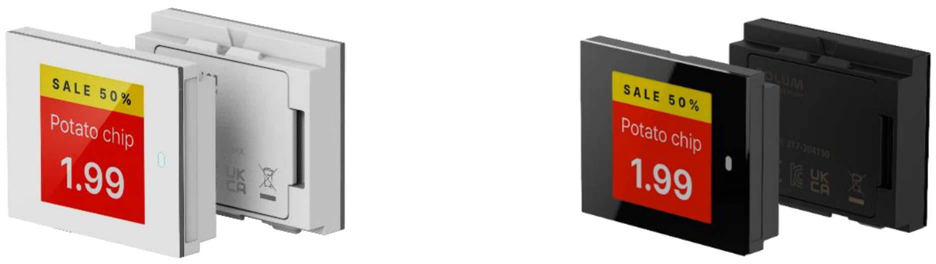
1.6” v
Model

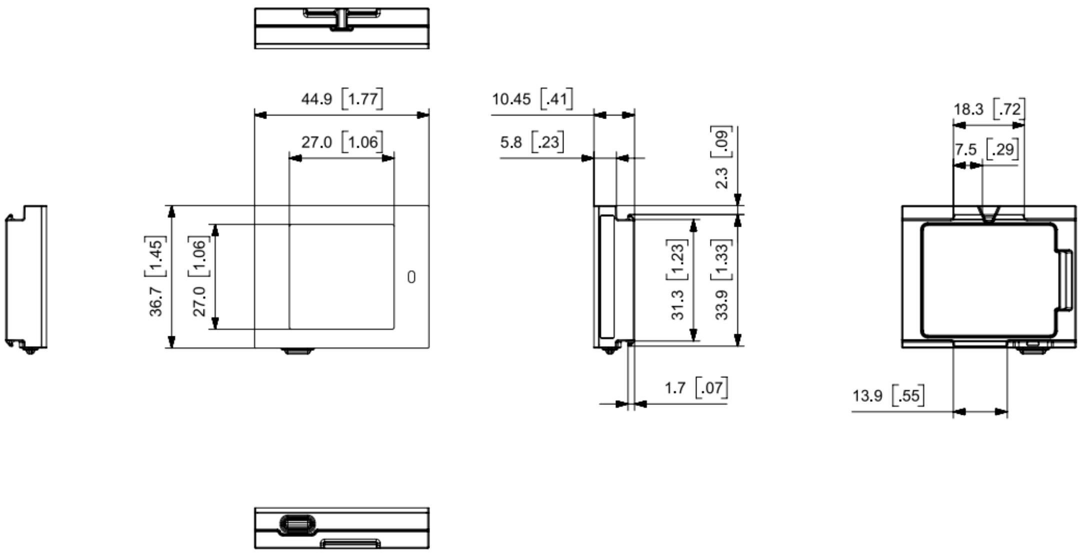


(Unit: mm)

Figure 3. 1.6” (V) Mechanical Dimension

1.6" H
Model

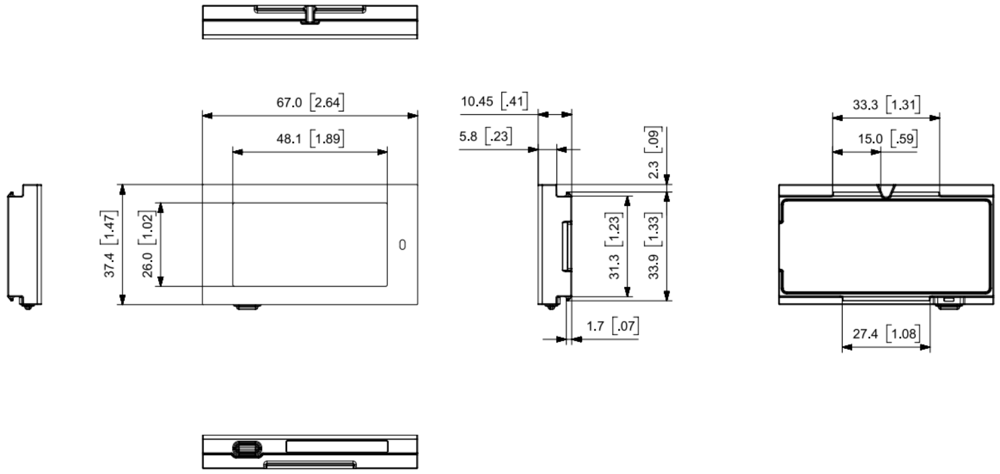




(Unit: mm)

Figure 4. 1.6" (H) Mechanical Dimension

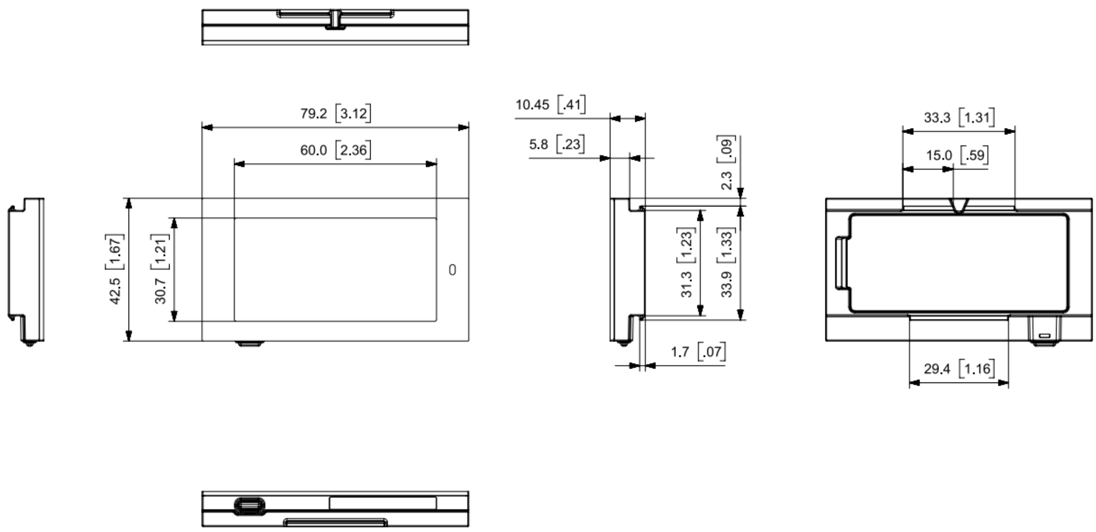
2.2"
Model



(Unit: mm)

Figure 5. 2.2" Mechanical Dimension

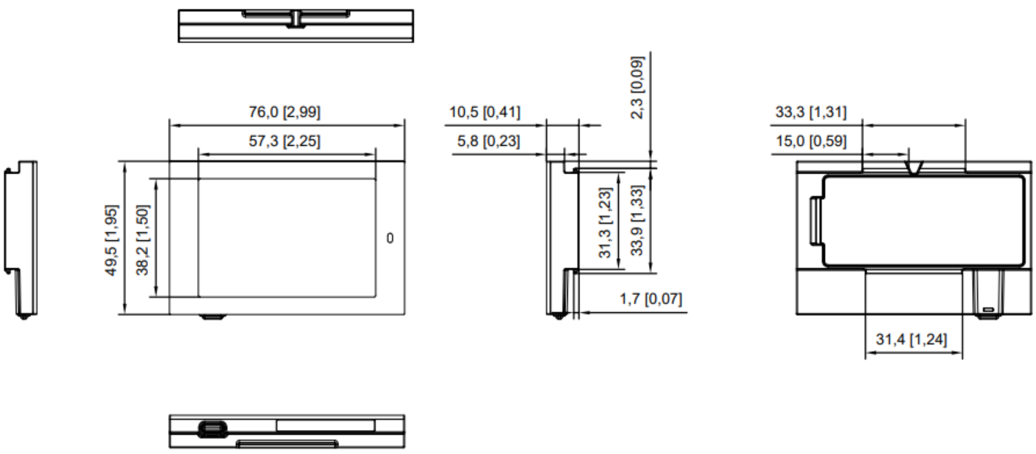
2.6"
Model



(Unit: mm)

Figure 6. 2.6" Mechanical Dimension

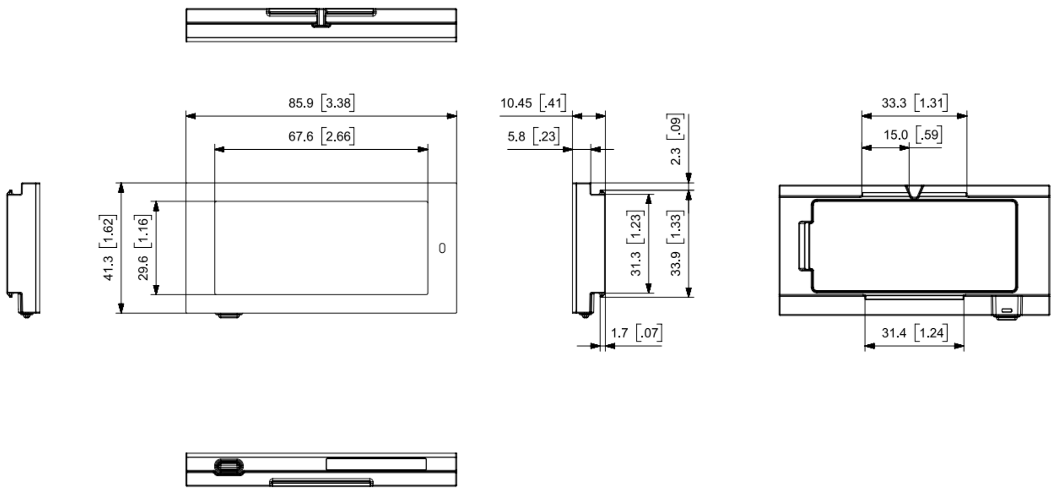
2.7”
Model



(Unit: mm)

Figure 7. 2.7” Mechanical Dimension

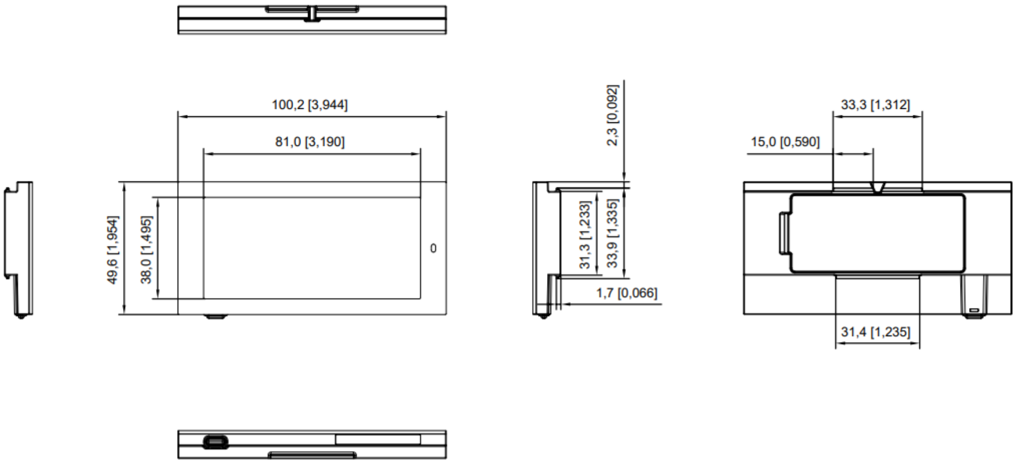
2.9”
Model



(Unit: mm, [inch])

Figure 8. 2.9” Mechanical Dimension

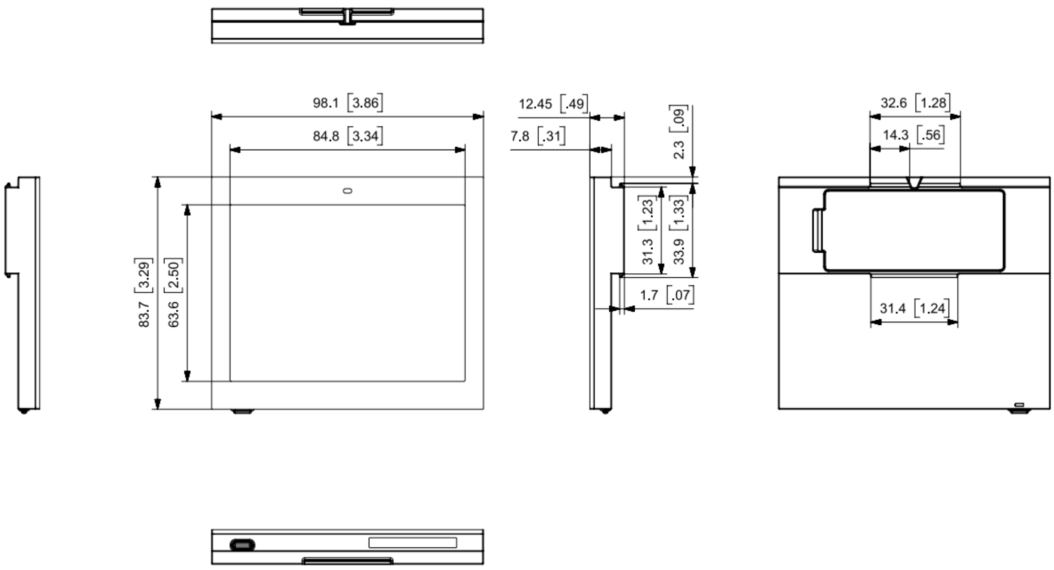
3.5"
Model



(Unit: mm, [inch])

Figure 9. 3.5" Mechanical Dimension

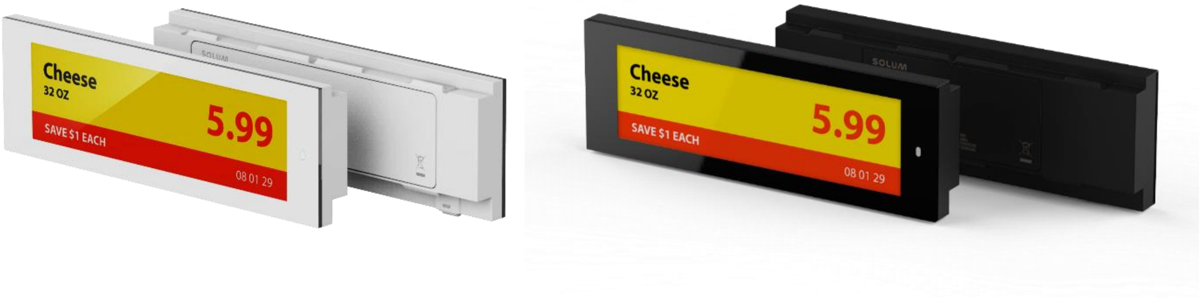
4.2”
Model

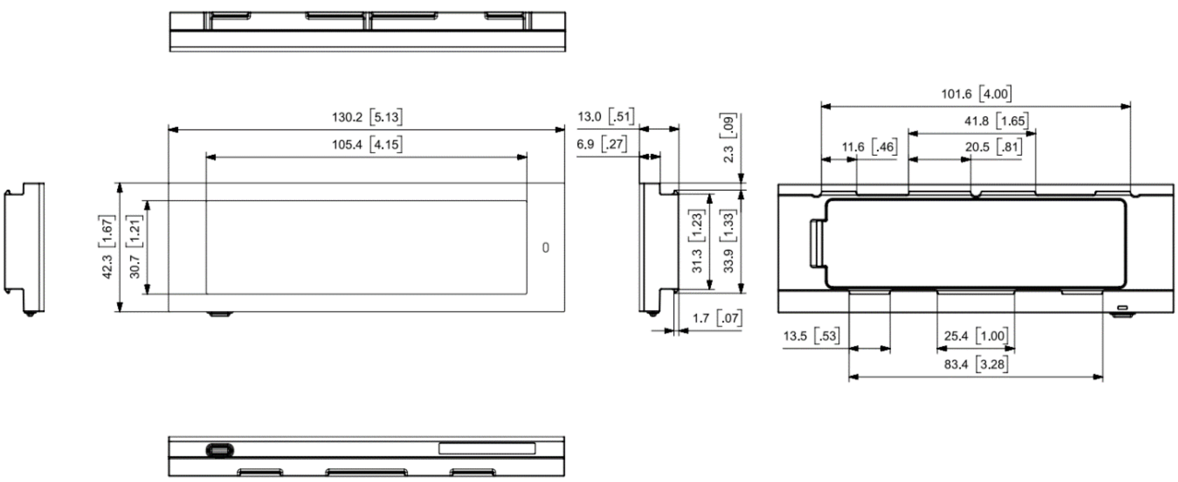


(Unit: mm, [inch])

Figure 10. 4.2” Mechanical Dimension

4.3"
Model

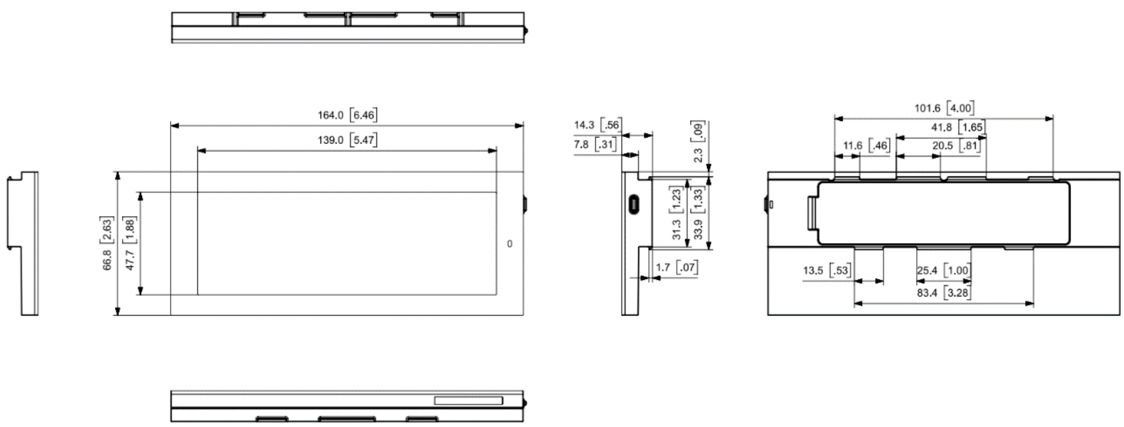




(Unit: mm, [inch])

Figure 11. 4.3" Mechanical Dimension

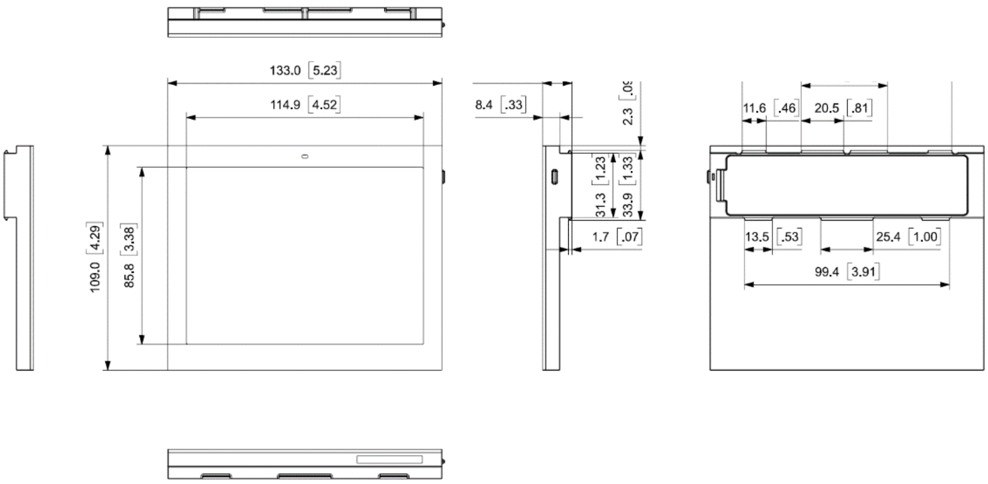
5.8”
Model



(Unit: mm, [inch])

Figure 12. 5.8” Mechanical Dimension

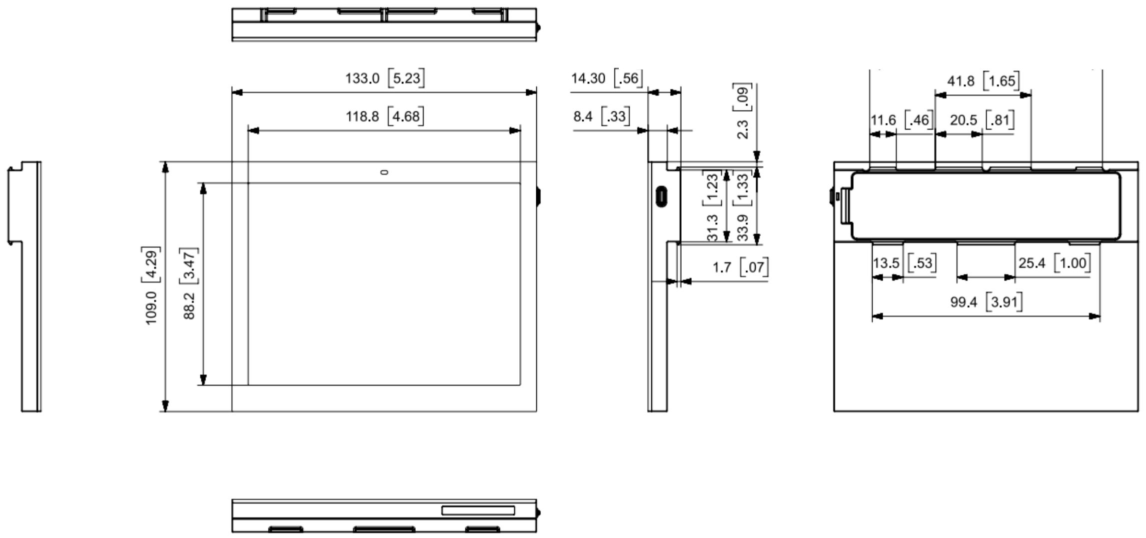
6.0”
Model



(Unit: mm, [inch])

Figure 13. 6.0” Mechanical Dimension

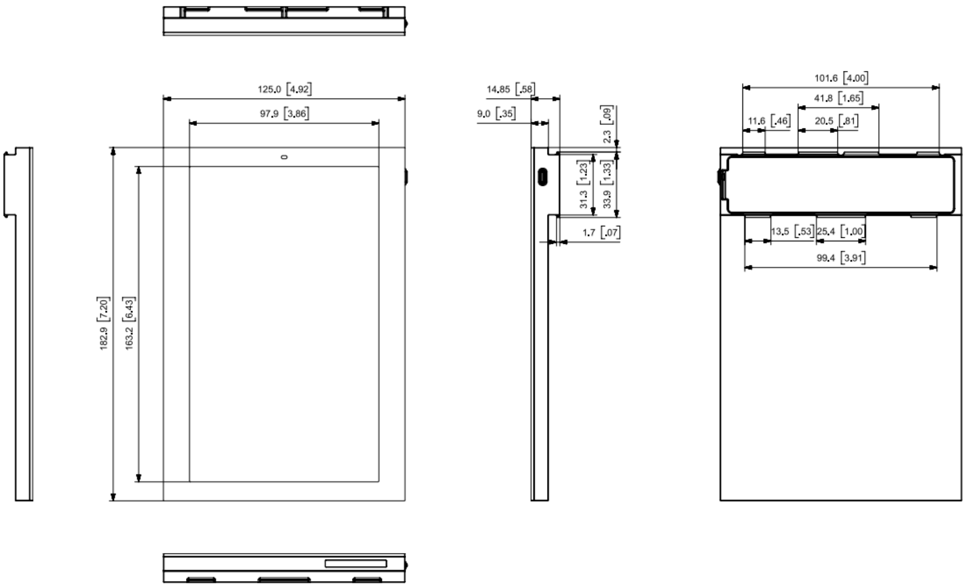
6.1” Model



(Unit: mm, [inch])

Figure 14. 6.1” Mechanical Dimension

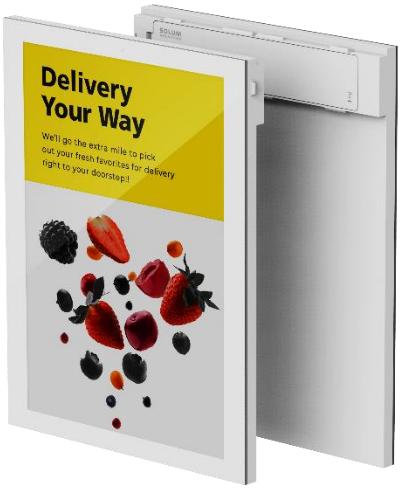
7.5”
Model

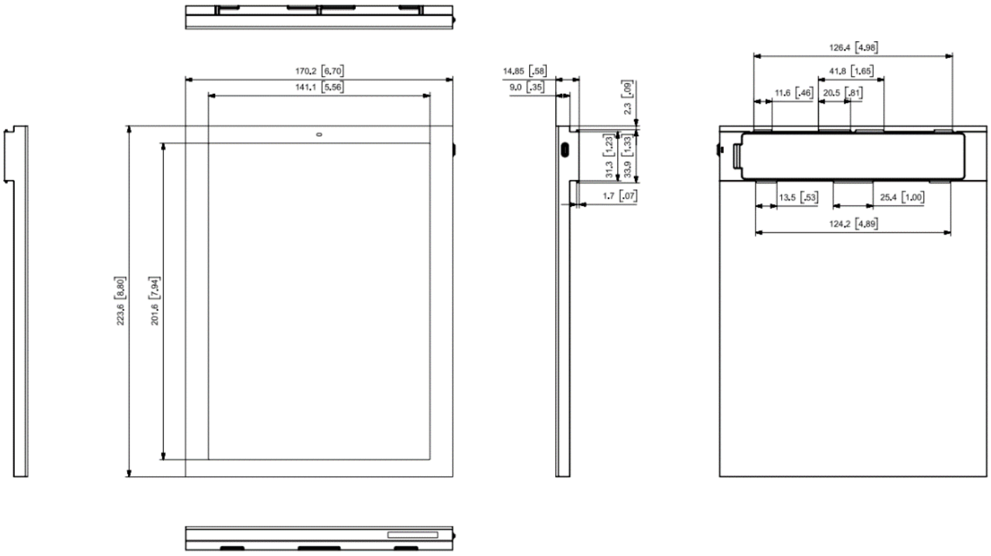


(Unit: mm, [inch])

Figure 15. 7.5” Mechanical Dimension

9.7"
Model

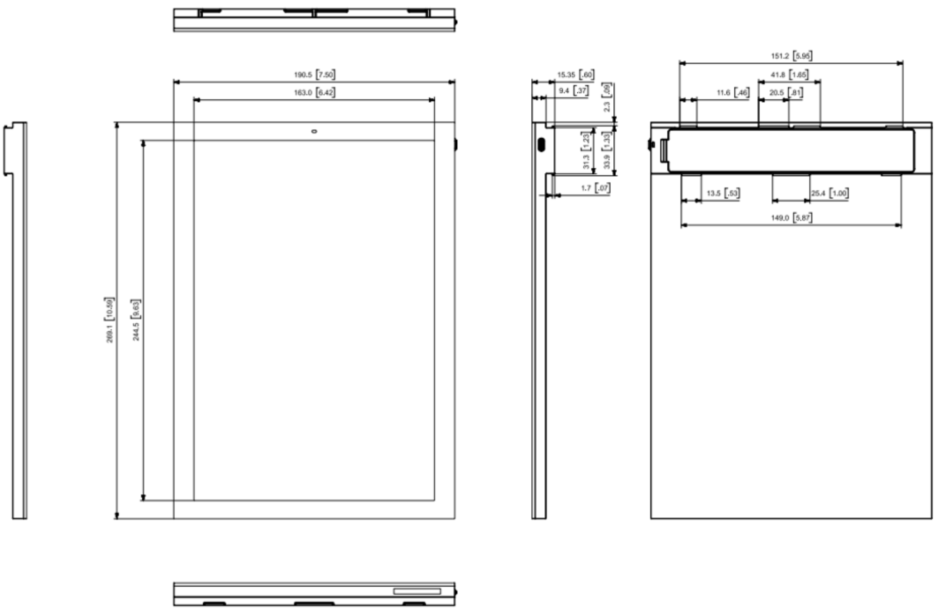




(Unit: mm, [inch])

Figure 16. 9.7" Mechanical Dimension

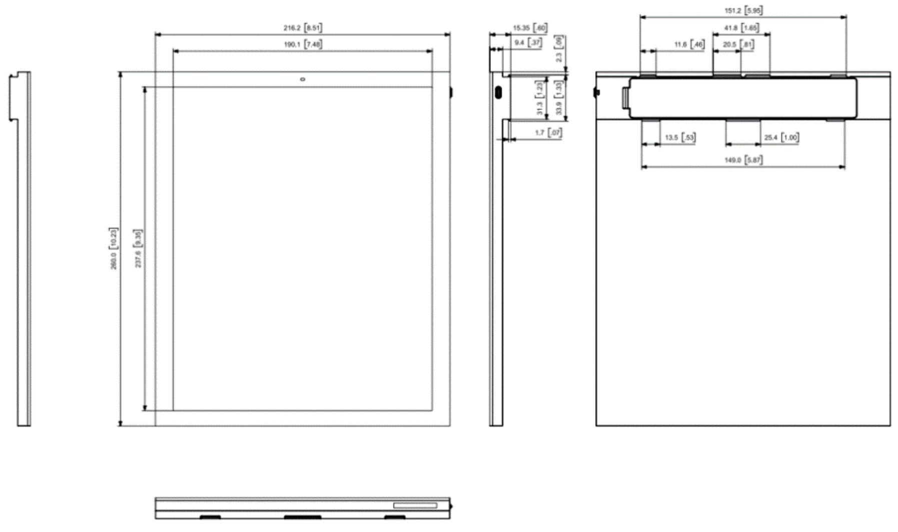
11.6”
Model



(Unit: mm, [inch])

Figure 17. 11.6” Mechanical Dimension

12.2”
Model



(Unit: mm, [inch])

Figure 18. 12.2” Mechanical Dimension

3.6 Label Marking

Figure 19. ESL Labels

ESL specific information can be found on the laser marking located on the back of the ESL.

Information displayed are : Model, Certifications

1) Model: EL-①②③**F6**④⑤⑥ (i.e. EL016F6W4A)
   - ① ② ③: ESL Display Size
     029=2.9”, 043=4.3”, 082=8.2”, 116=11.6”
   - ④: Housing Color
     W =white, B =black
   - ⑤: ESL Display Color
     4= RED, YELLOW (BWRY)
   - ⑥: ESL Tag Type
     A=(Button, NFC,LED)

Size	Model Name
1.6V”	EL016F6W4A
1.6H”	EL016F6B4A
2.2"	EL022F6W4A
2.6"	EL026F6W4A
2.7”	EL027F6B4A
2.9"	EL029F6W4A
3.5”	EL035F6W4A
4.2”	EL042F6W4A
4.3”	EL043F6W4A
5.8”	EL058F6W4A
6.0”	EL060F6W4A
6.1”	EL060H6W4A
7.5”	EL075F6W4A
9.7”	EL097F6W4A
11.6”	EL116F6W4A
12.2”	EL122F6W4A

3.7 Initial display image

ESL specific information can be found on the image of the ESL. Information displayed are but not limited to : IC, FCC ID, Model, ESL Media Access and Control (MAC) Address.

Information can be found in the image below. MAC barcode and 12-digits code will be shown on the display and the label can be found in the side of the ESL as well. They are to be scanned for operation.

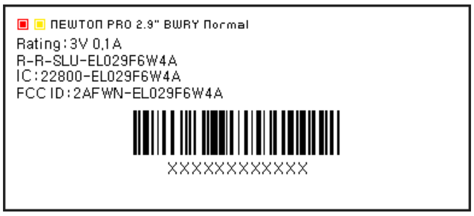


Figure 20. ESL Labels (MAC)

3.8 Certifications

Each ESL has the following certifications and should be noted for your region.

Size	Model	FCC	IC	CE*	TELEC	KC
1.6V”	EL016F6W4A	o	o	o	o	o
1.6H”	EL016F6B4A	o	o	o	o	o
2.2"	EL022F6W4A	o	o	o	o	o
2.6"	EL026F6W4A	o	o	o	o	o
2.7”	EL027F6B4A	o	o	o	o	o
2.9"	EL029F6W4A	o	o	o	o	o
3.5”	EL035F6W4A	o	o	o	o	o
4.2”	EL042F6W4A	o	o	o	o	o
4.3”	EL043F6W4A	o	o	o	o	o
5.8”	EL058F6W4A	o	o	o	o	o
6.0”	EL060F6W4A	o	o	o	o	o
6.1”	EL060H6W4A	o	o	o	o	o
7.5”	EL075F6W4A	o	o	o	o	o
9.7”	EL097F6W4A	o	o	o	o	o
11.6”	EL116F6W4A	o	o	o	o	o
12.2”	EL122F6W4A	o	o	o	o	o

- IEC 62471 standard (Photobiological Safety of Lamps and Lamp Systems) compliant

***CE: RF/EMC/LVD/RoHS2**

3.8.1 FCC

WARNING: This equipment may generate or use radio frequency energy. Changes or modifications to this equipment may cause harmful interference unless the modifications are expressly approved in the instruction manual. The user could lose the authority to operate this equipment if an unauthorized change or modification is made.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications, However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

3.8.2 CE

We hereby declare under our sole responsibility that the electrical product above is in compliance with the essential requirements of the Radio Equipment Directive (2014/53/EU) by application of
EN 60950-1:2006+A11:2009+A1:2010+A12:2011+A2:2013
EN 62479:2010
EN 301 489-1 V2.2.0
EN 301 489-17 V3.2.0
EN 300 328 V2.2.2
and the Directive (2011/65/EU) on the restriction of the use of certain hazardous substances in electrical and electronic equipment by application of EN 62321 Series.

3.8.3 IC

This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

4 Packaging

4.1 1.6" (V)

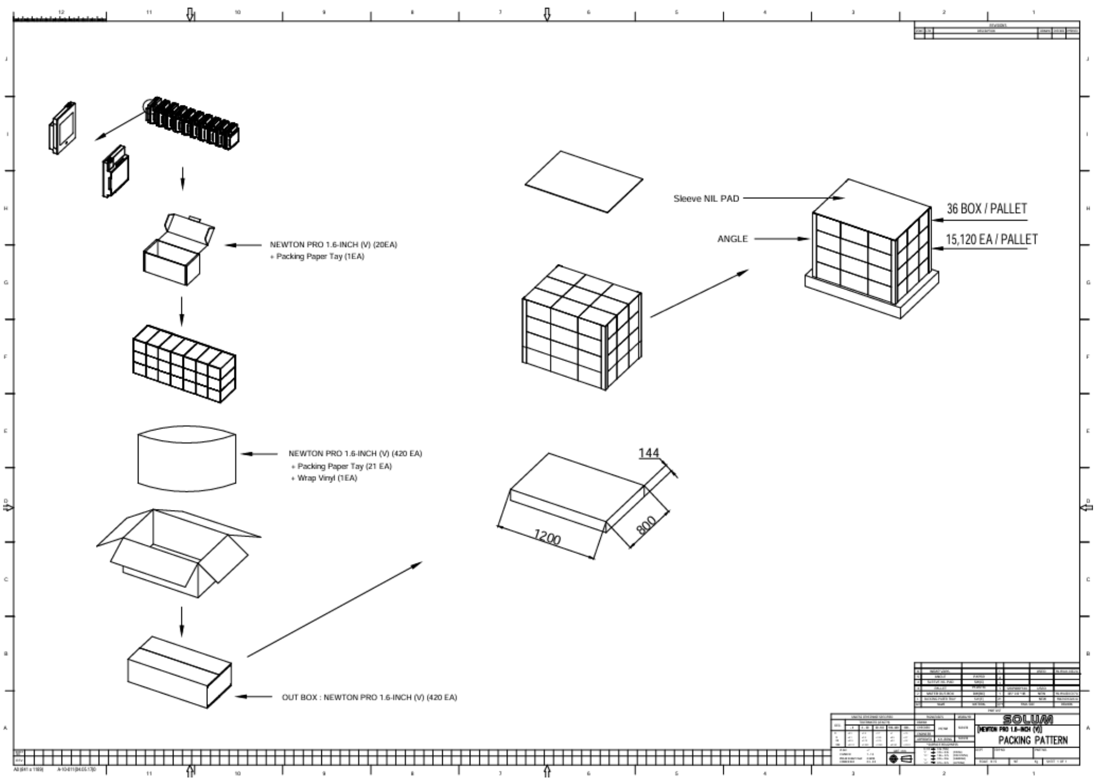


Figure 21. 1.6"(V) Packing-Diagram (1,200 X 800)

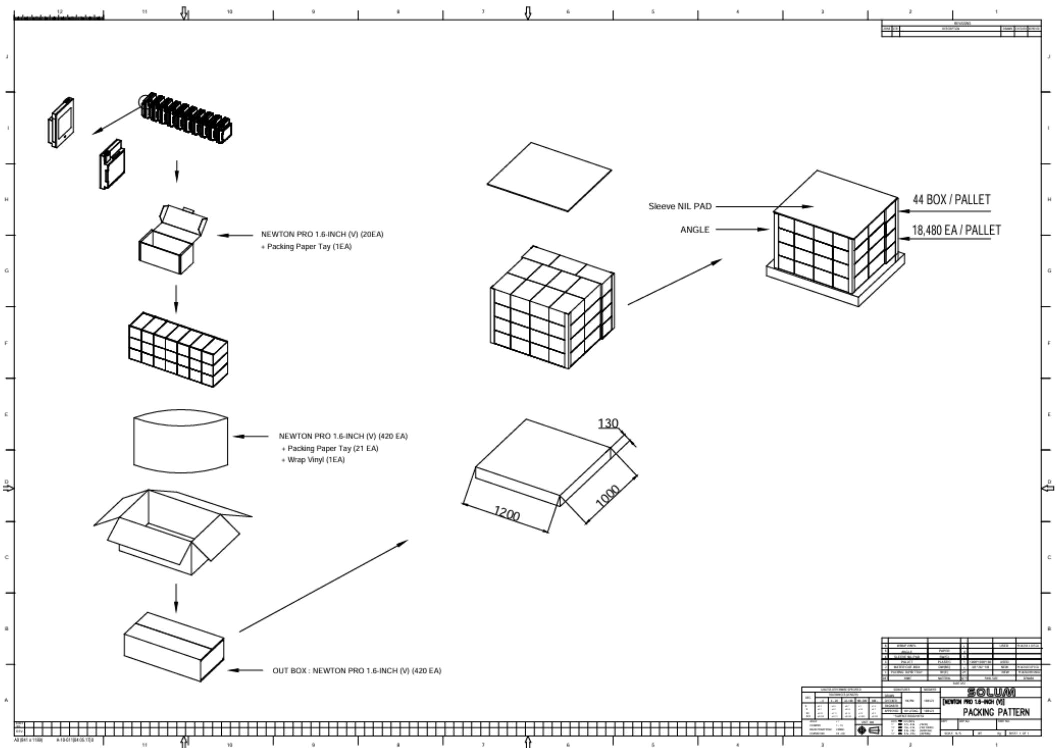


Figure 22. 1.6"(V) Packing-Diagram (1,200 X 1,000)

4.2 1.6” (H)

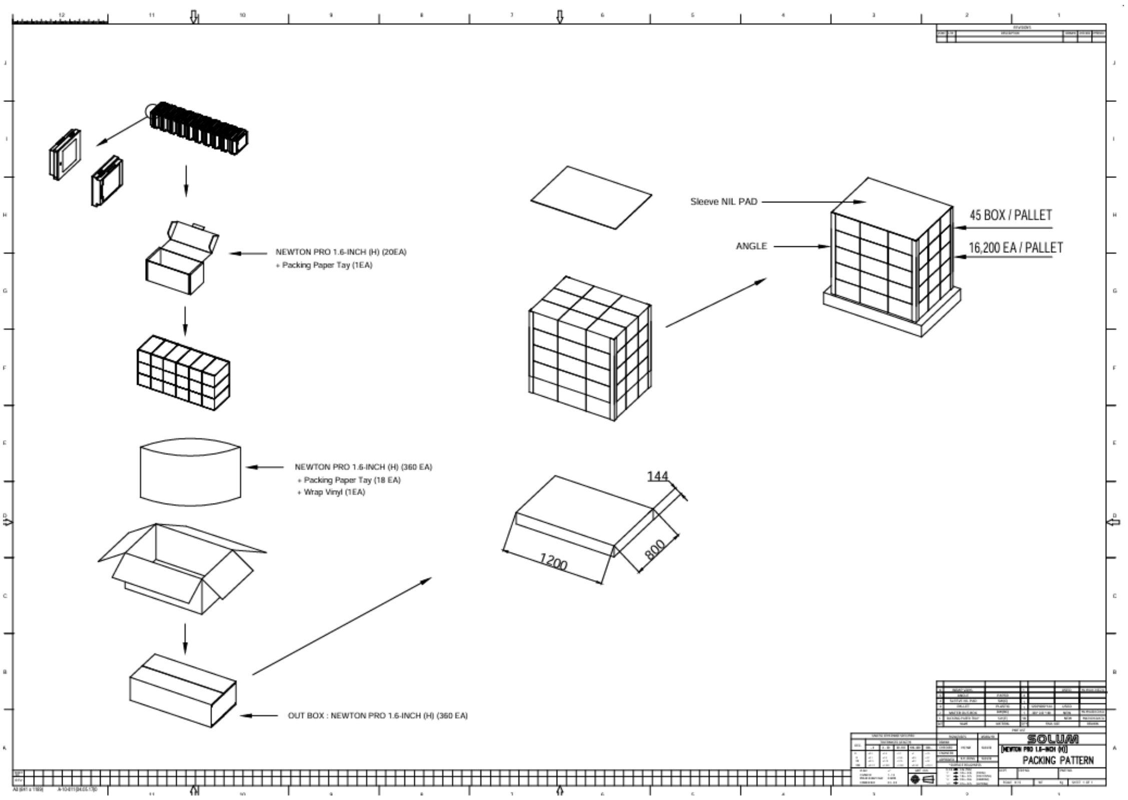


Figure 23. 1.6”(H) Packing-Diagram (1,200 X 800)

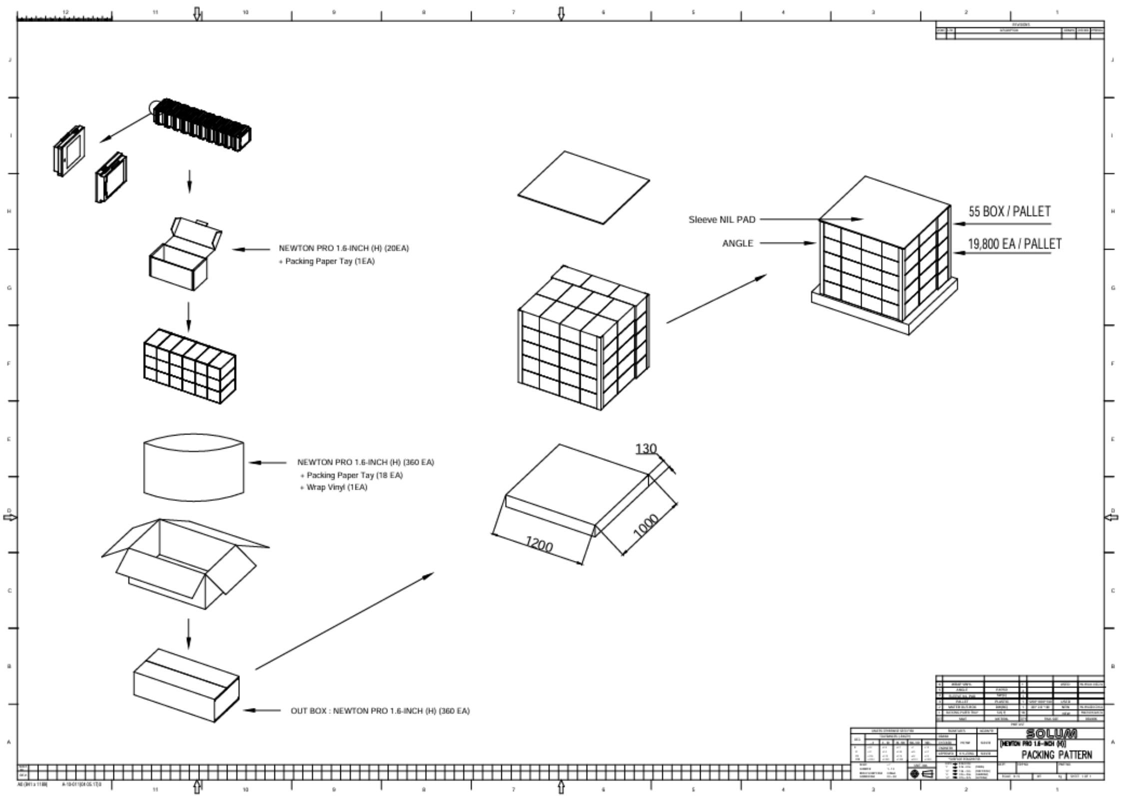


Figure 24. 1.6”(H) Packing-Diagram (1,200 X 1,000)

4.3 2.2”

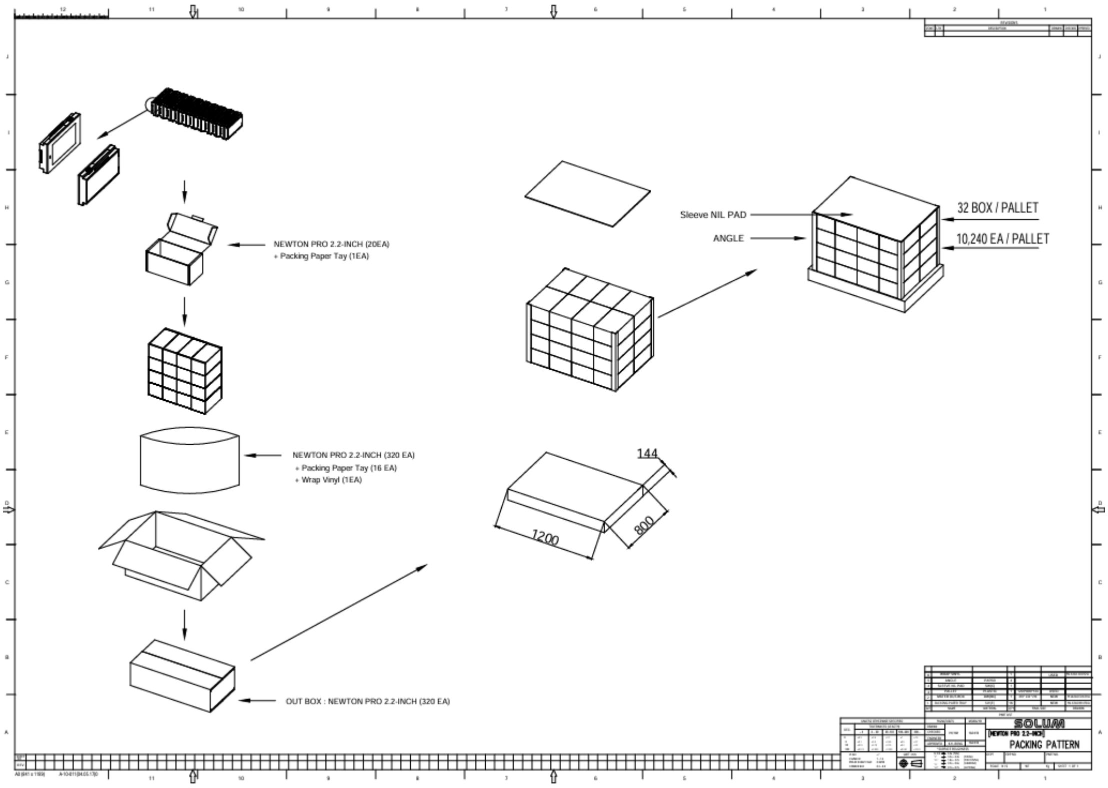


Figure 25. 2.2” Packing-Diagram (1,200 X 800)

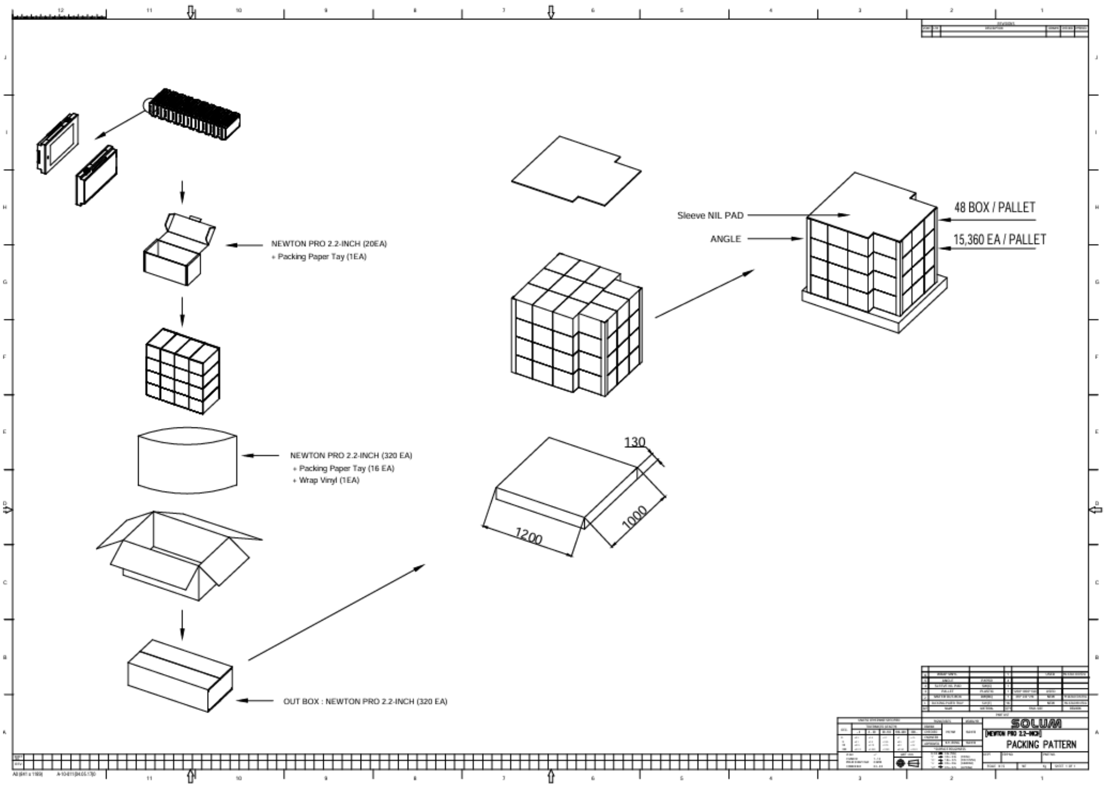


Figure 26. 2.2” Packing-Diagram (1,200 X 1,000)

4.4 2.6”

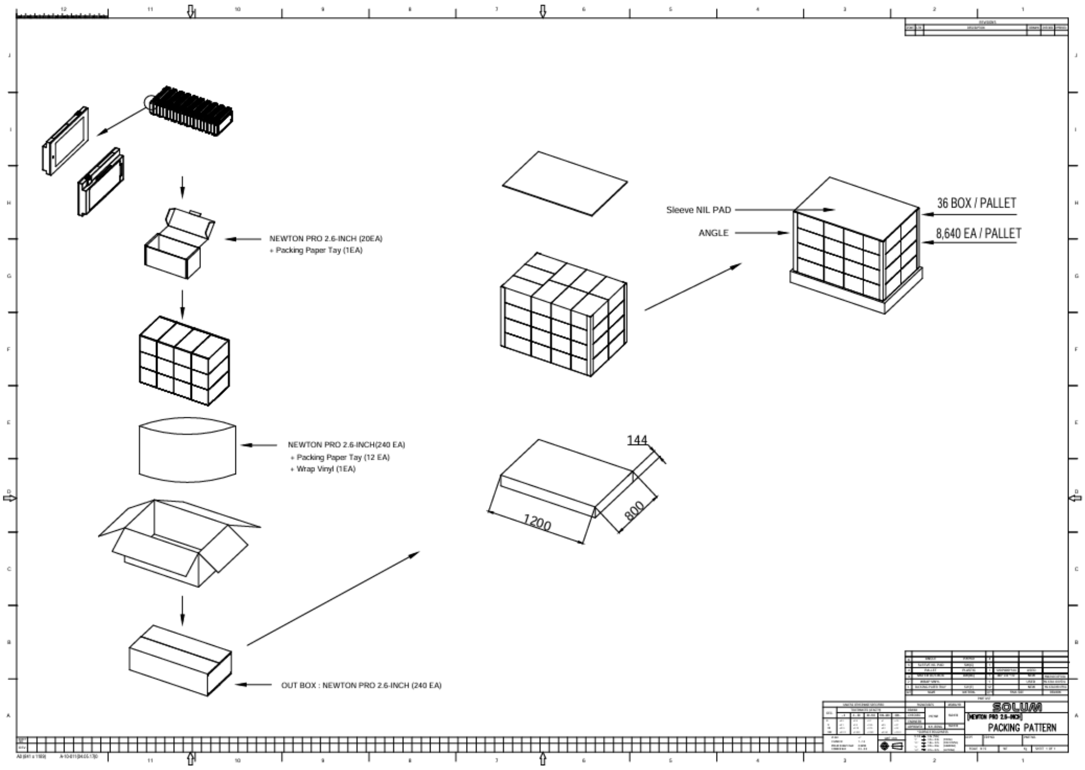


Figure 27. 2.6” Packing-Diagram (1,200 X 800)

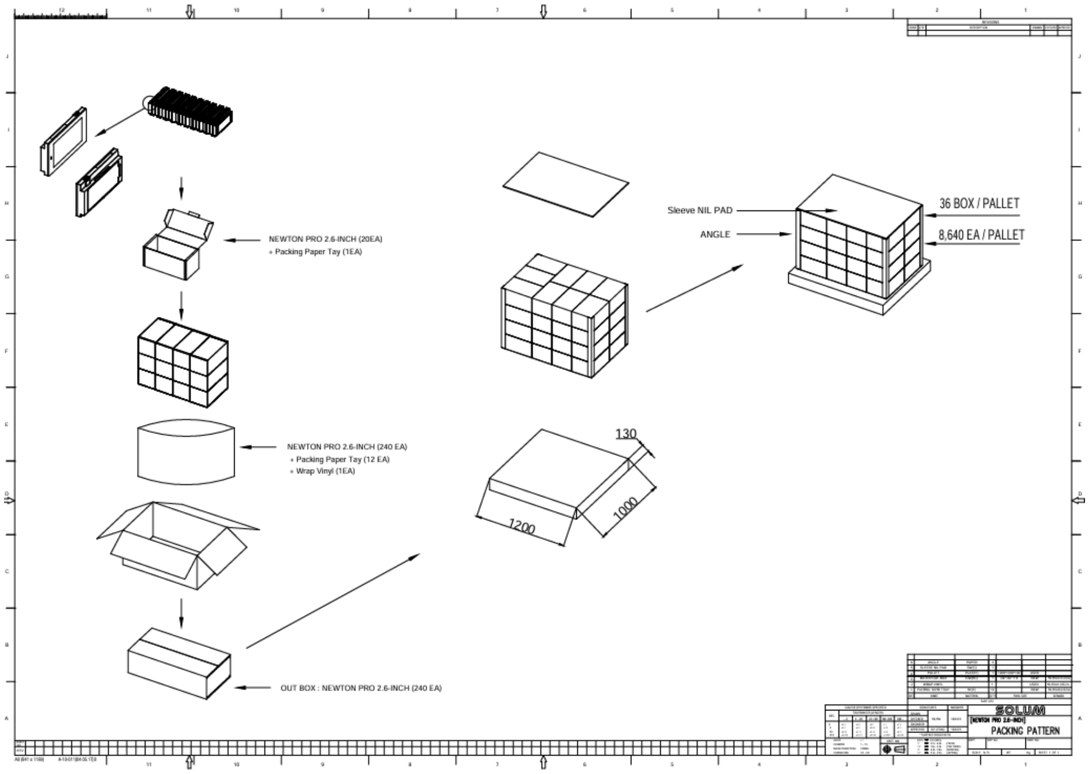


Figure 28. 2.6” Packing-Diagram (1,200 X 1,000)

4.5 2.7”

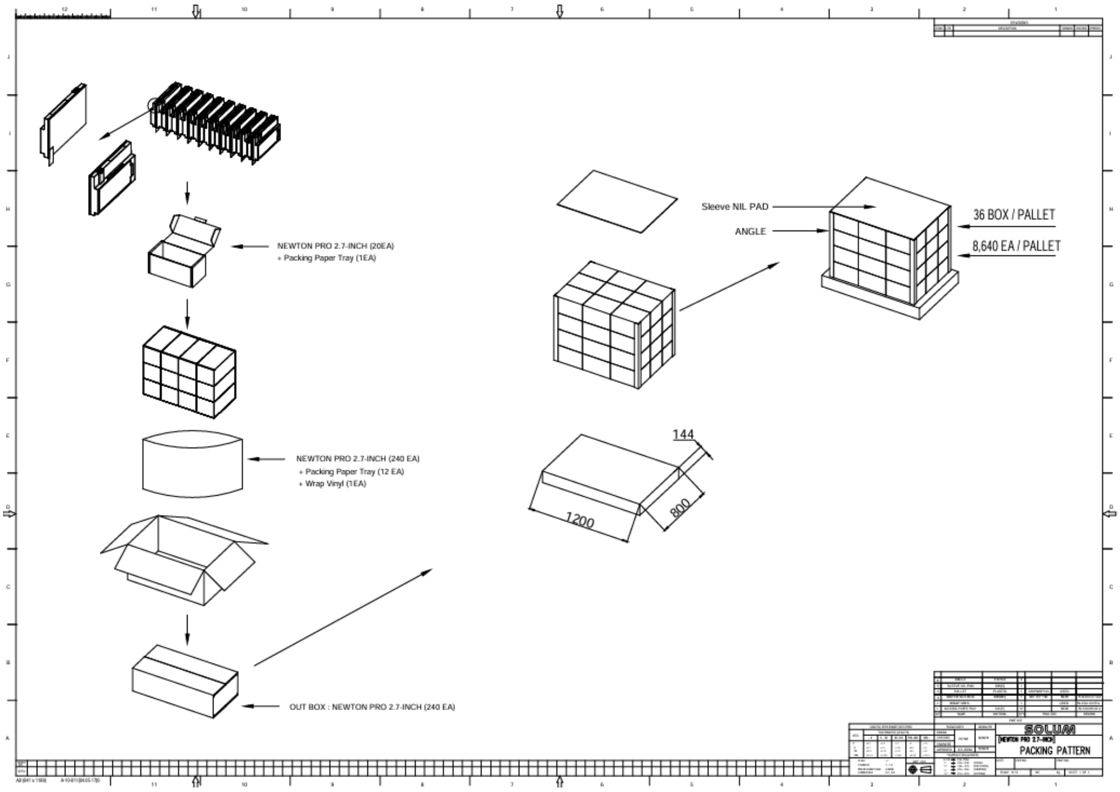


Figure 29. 2.7” Packing-Diagram (1,200 X 800)

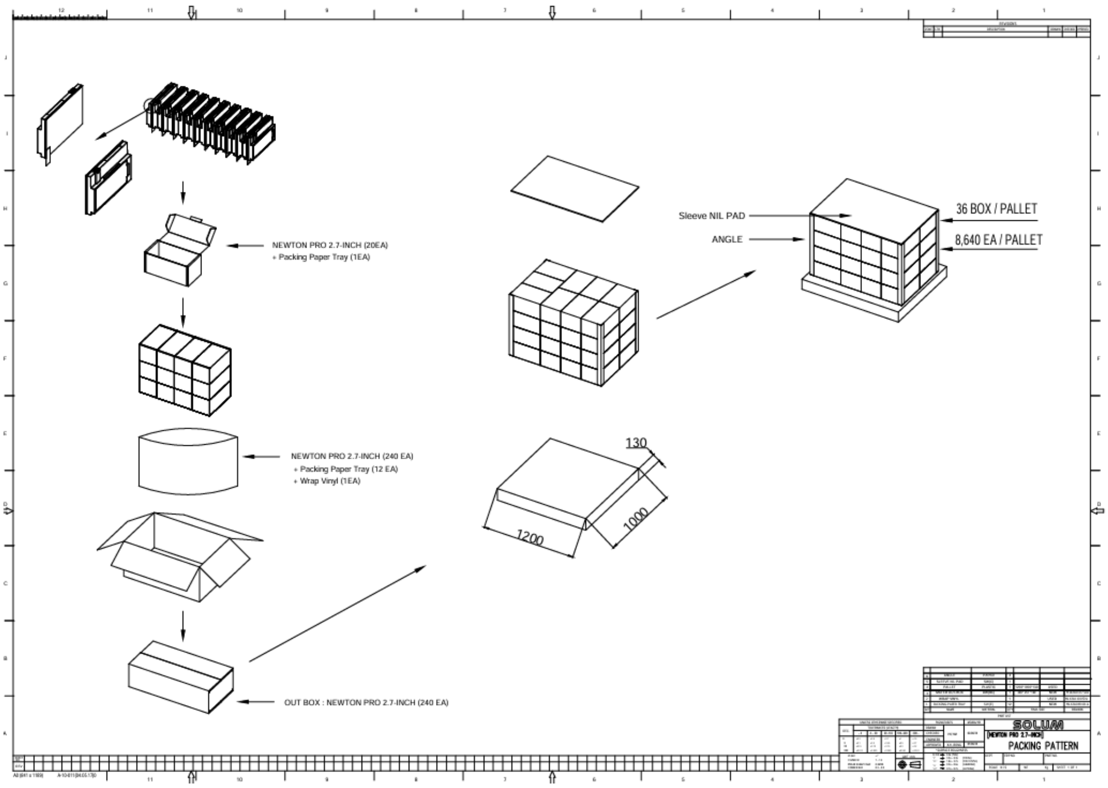


Figure 30. 2.7” Packing-Diagram (1,200 X 1,000)

4.6 2.9”

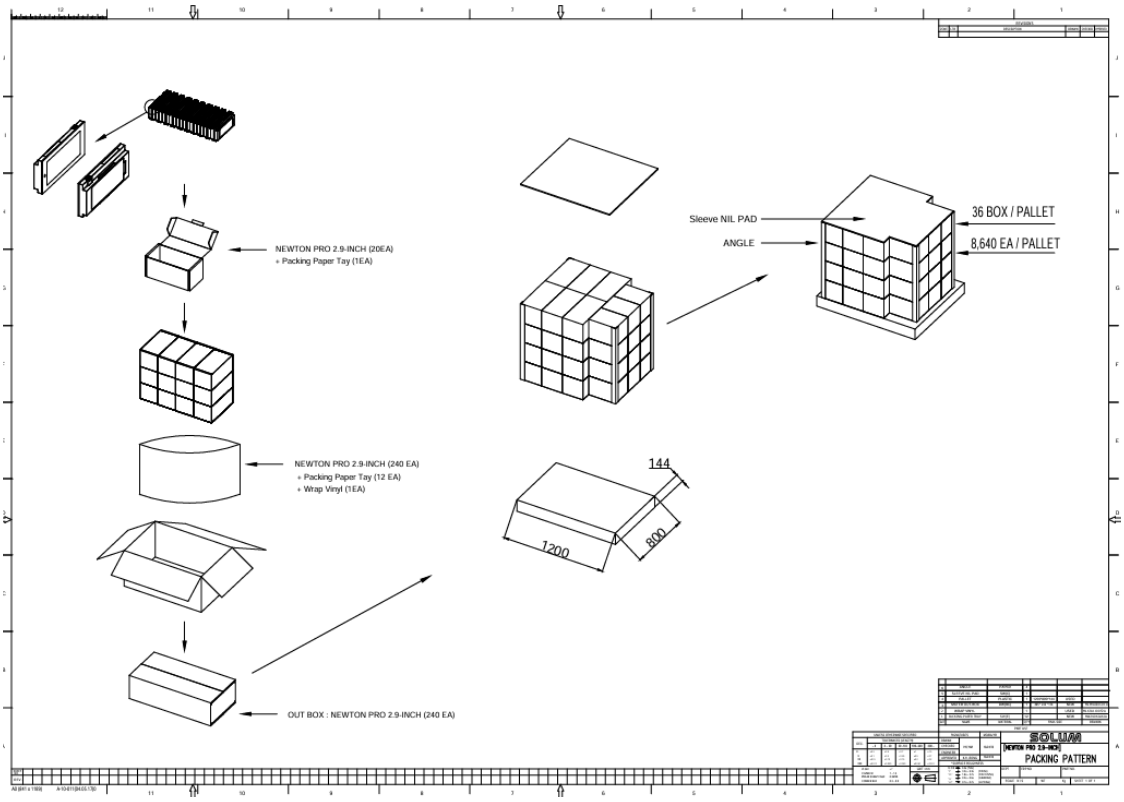


Figure 31. 2.9” Packing-Diagram (1,200 X 800)

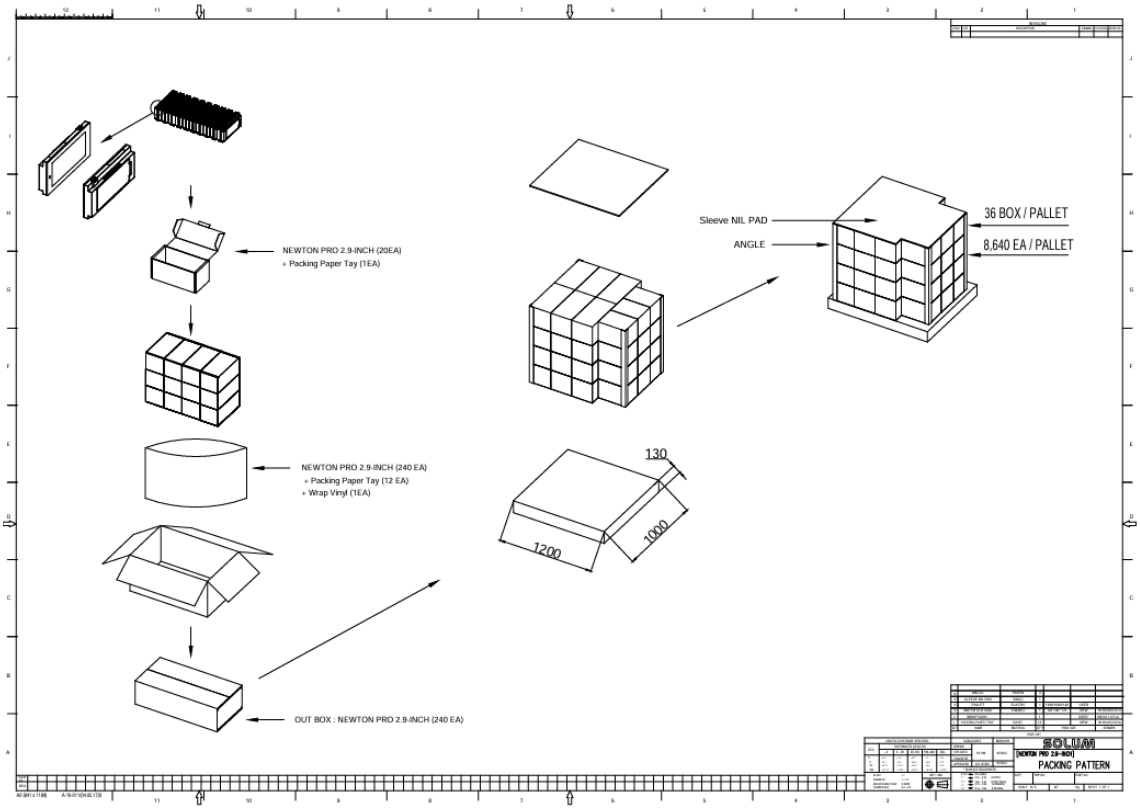


Figure 32. 2.9” Packing-Diagram (1,200 X 1,000)

4.7 3.5”

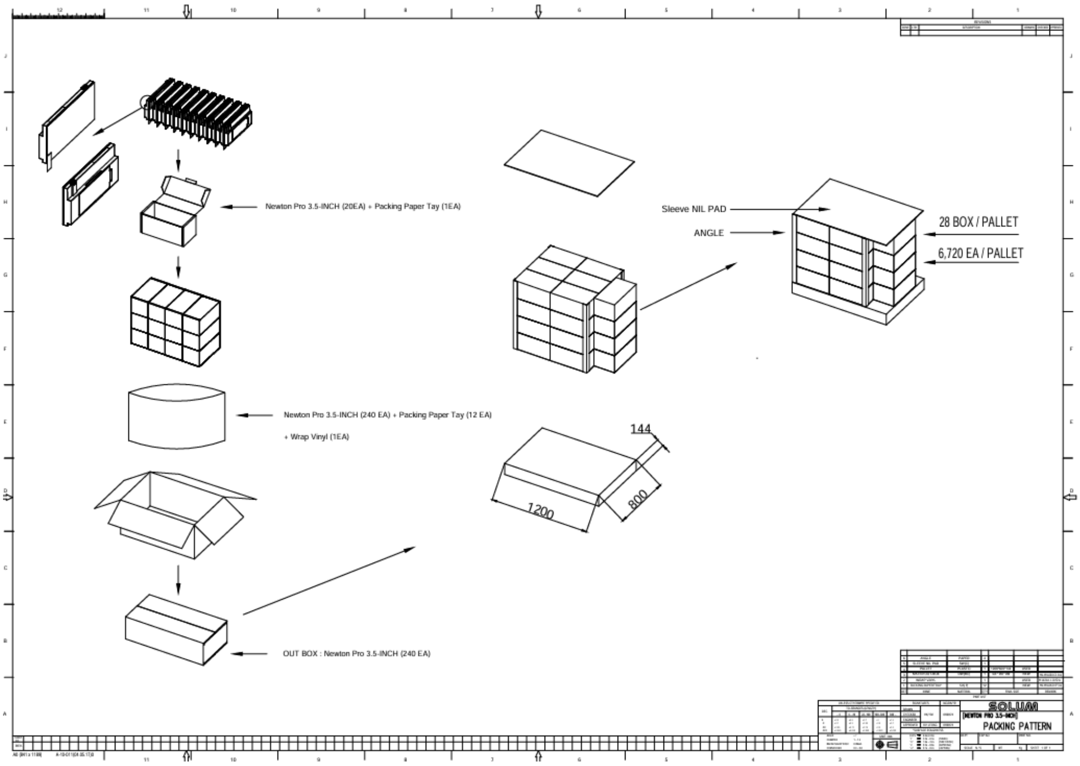


Figure 33. 3.5” Packing-Diagram (1,200 X 800)

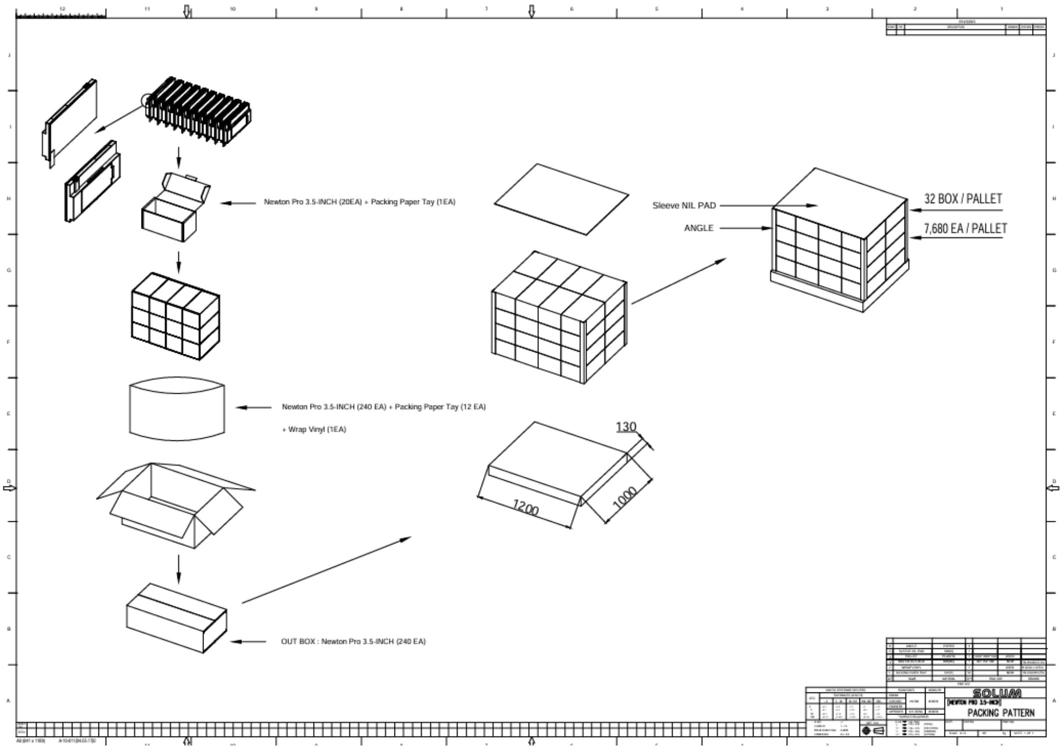


Figure 34. 3.5” Packing-Diagram (1,200 X 1,000)

4.8 4.2”

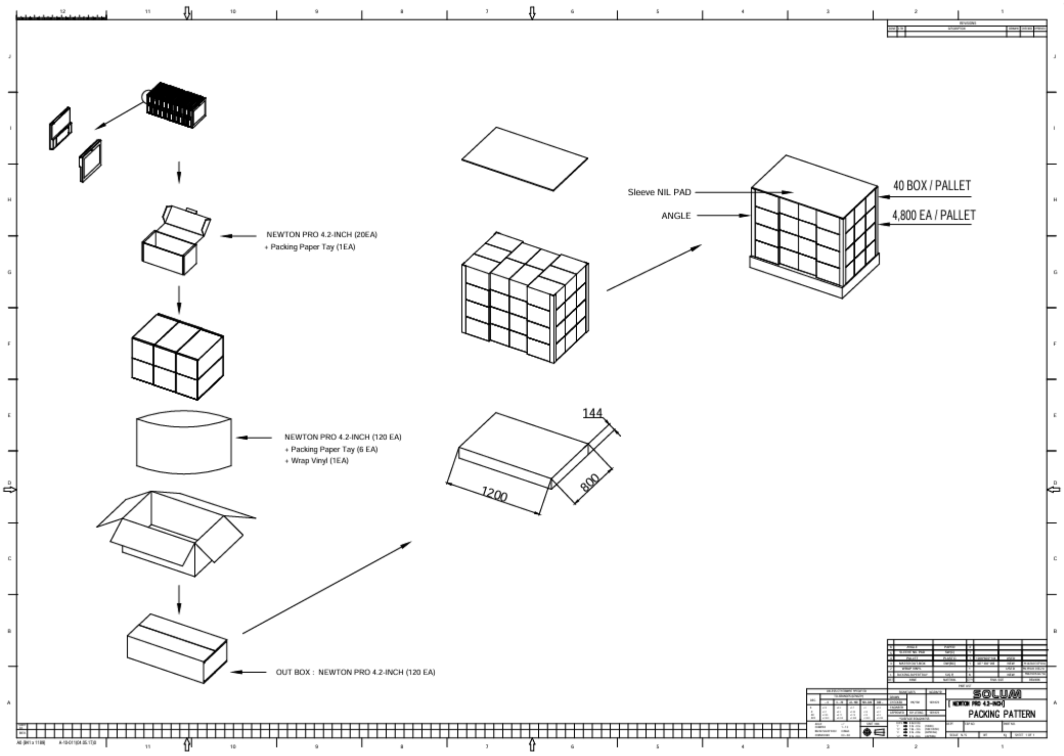


Figure 35. 4.2” Packing-Diagram (1,200 X 800)

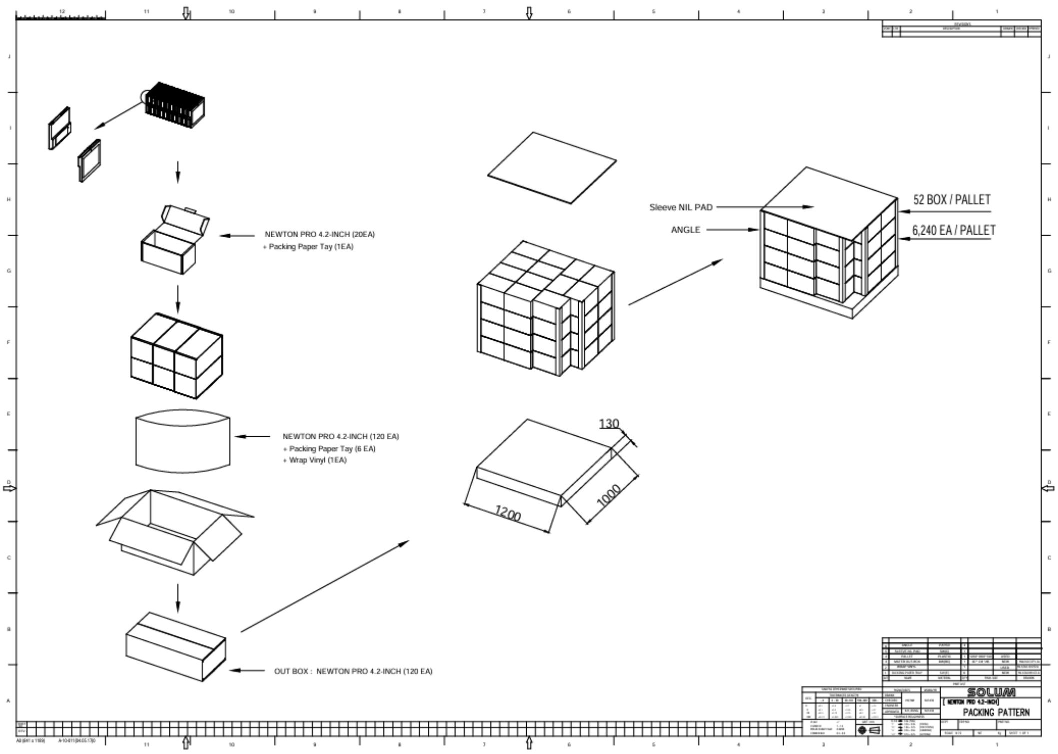


Figure 36. 4.2” Packing-Diagram (1,200 X 1,000)

4.9 4.3”

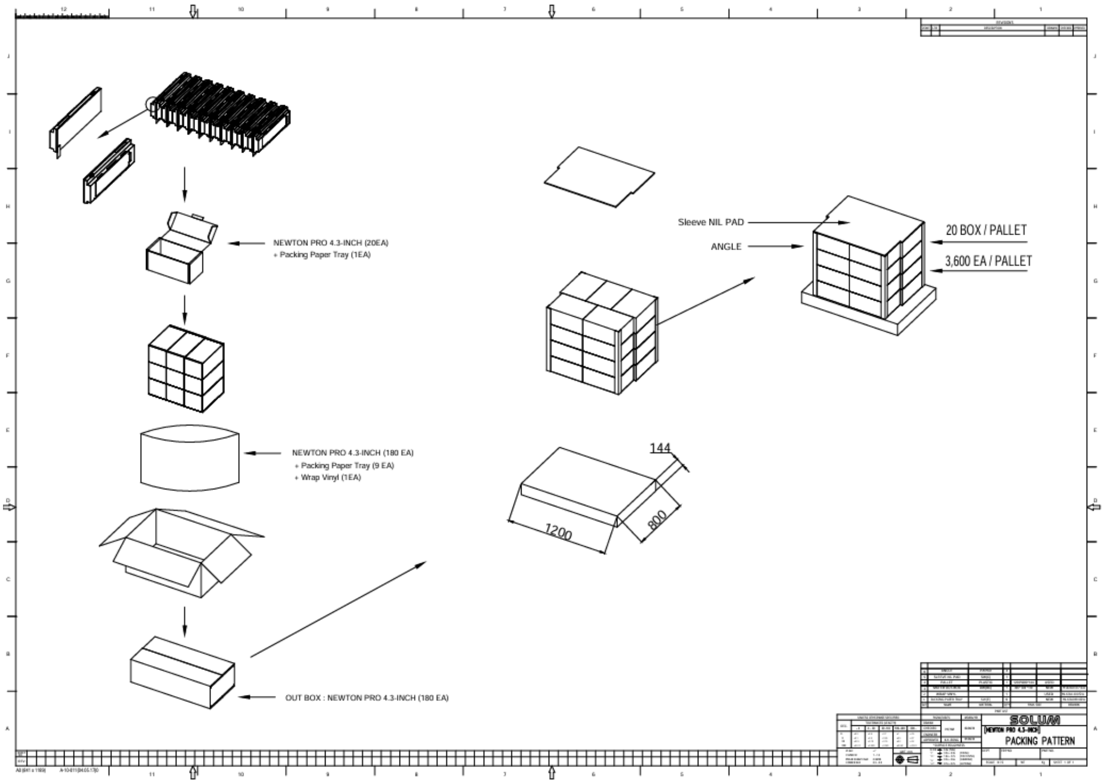


Figure 37. 4.3” Packing-Diagram (1,200 X 800)

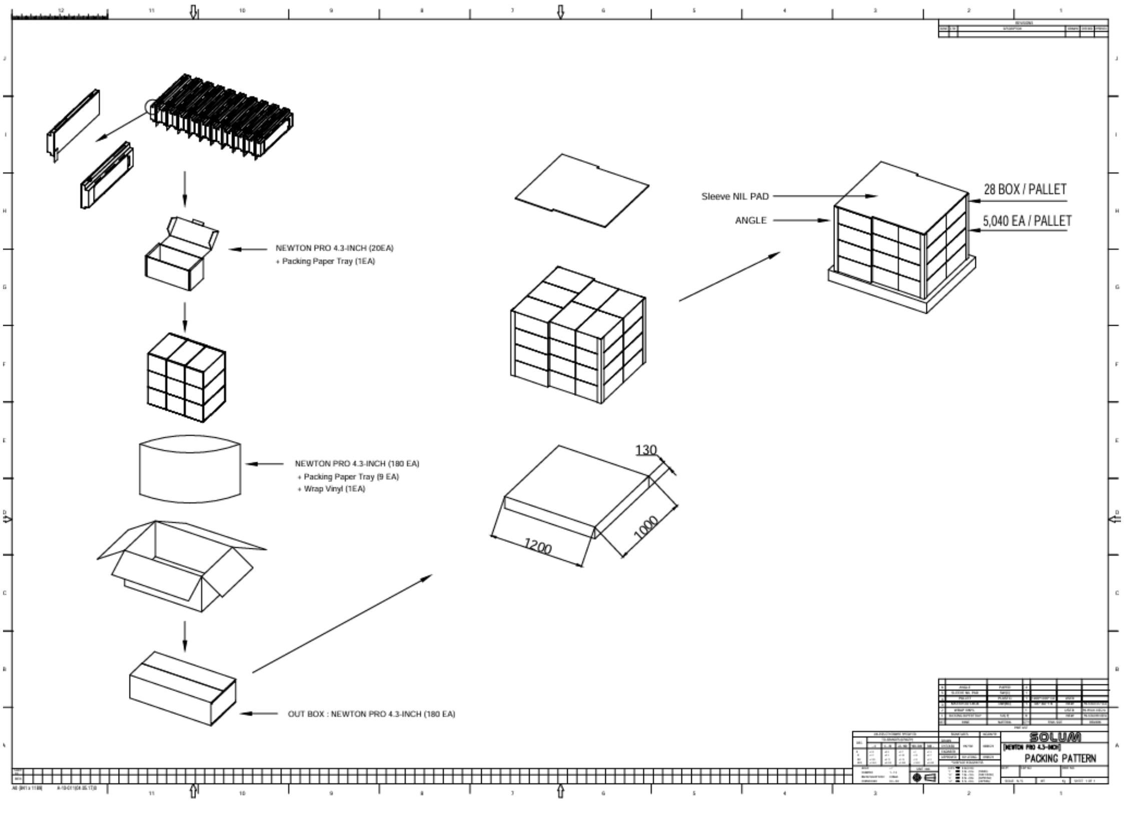


Figure 38. 4.3” Packing-Diagram (1,200 X 1,000)

4.10 6.0"

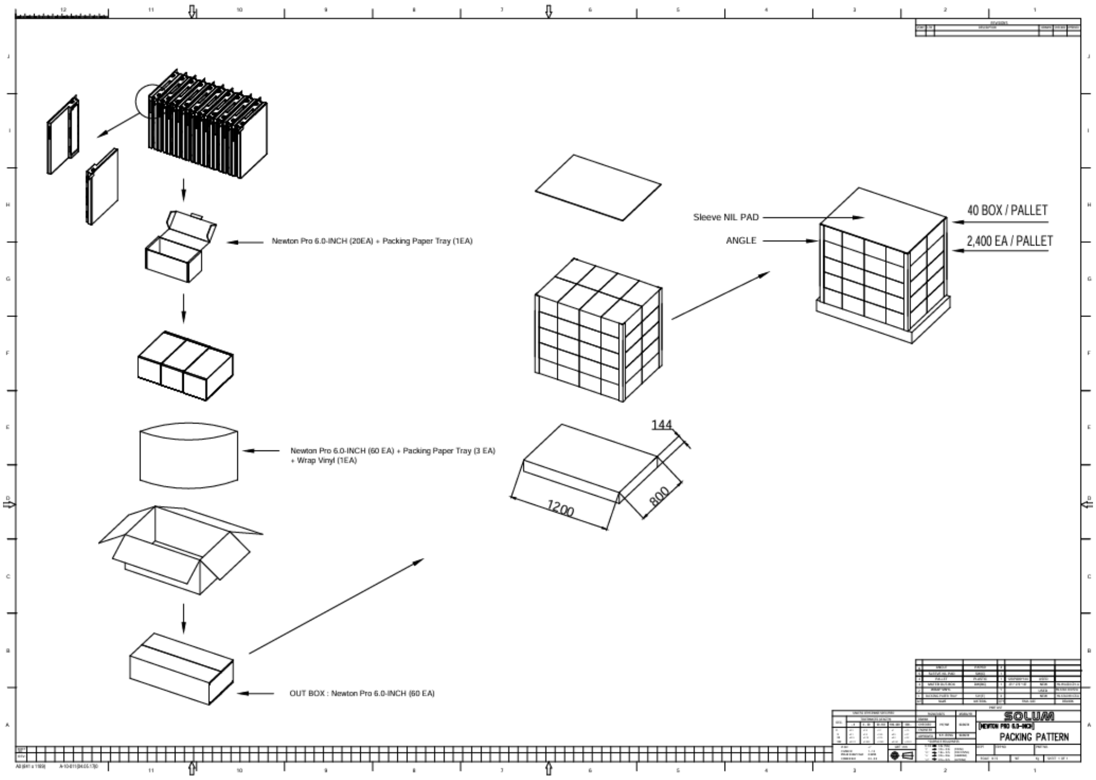


Figure 39. 6.0" Packing-Diagram (1,200 X 800)

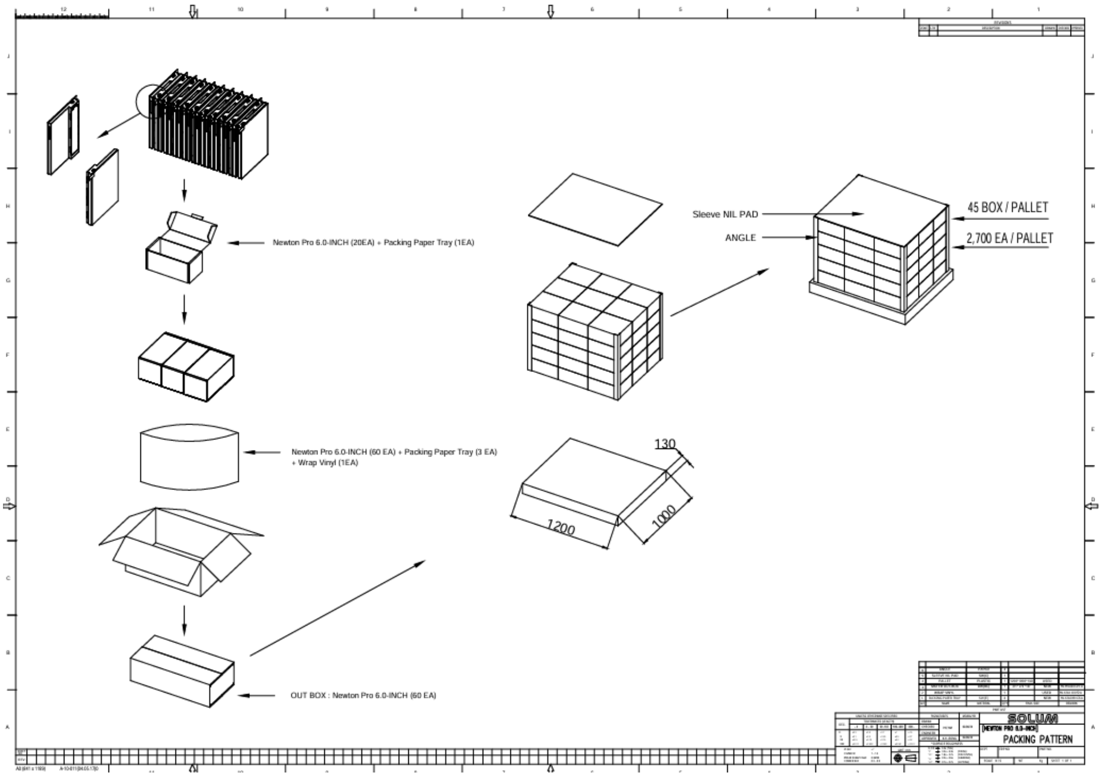


Figure 40. 6.0" Packing-Diagram (1,200 X 1,000)

4.11 6.1"

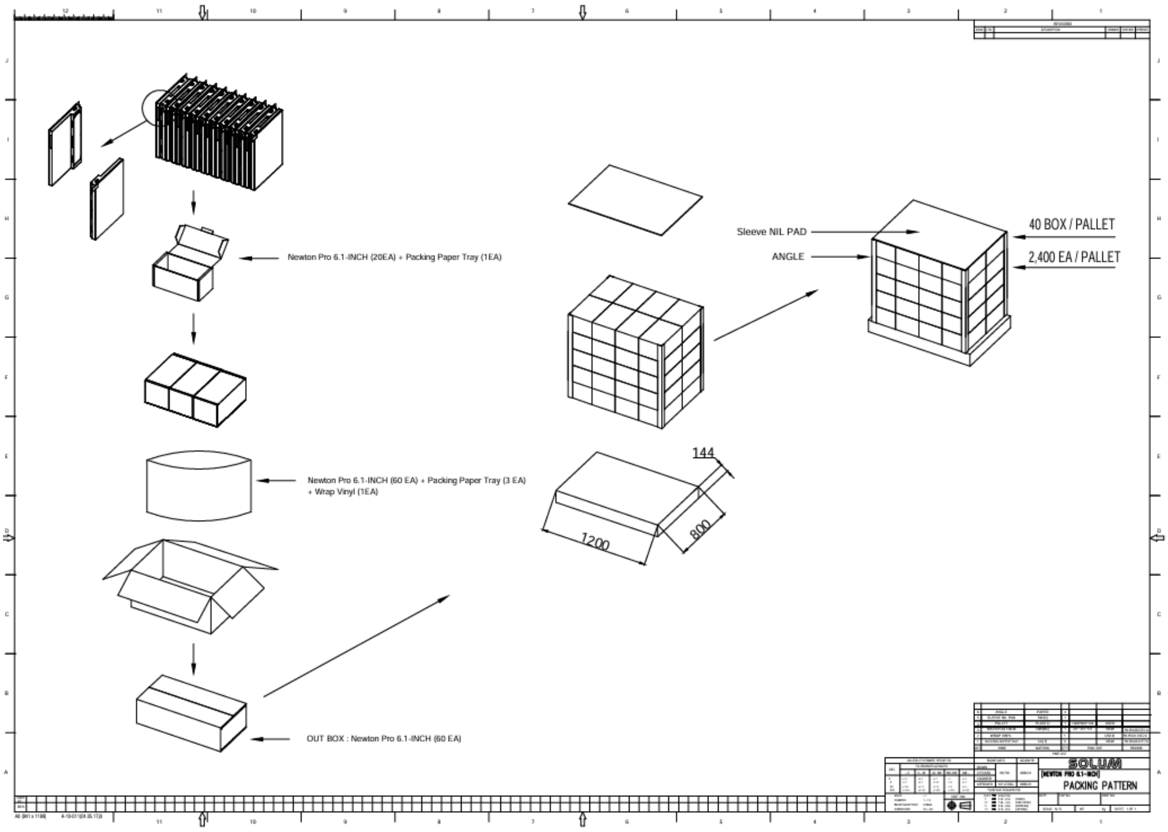


Figure 41. 6.1" Packing-Diagram (1,200 X 800)

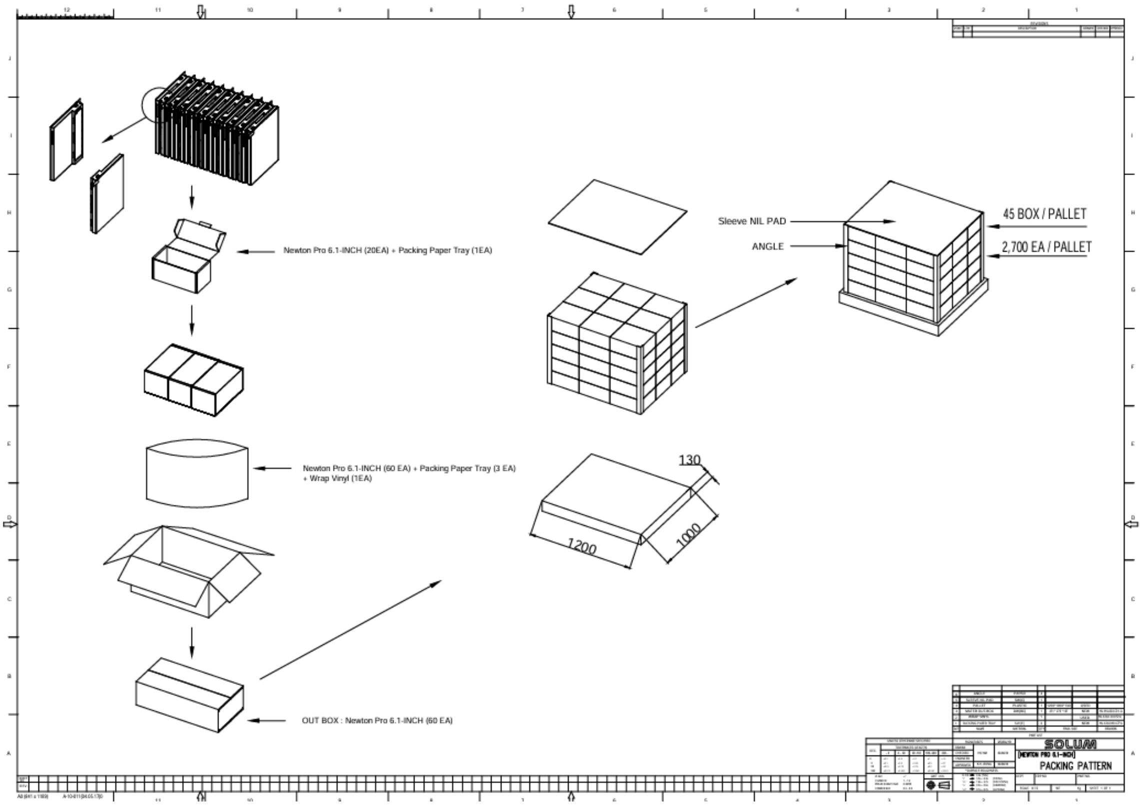


Figure 42. 6.1" Packing-Diagram (1,200 X 1,000)

4.12 7.5”

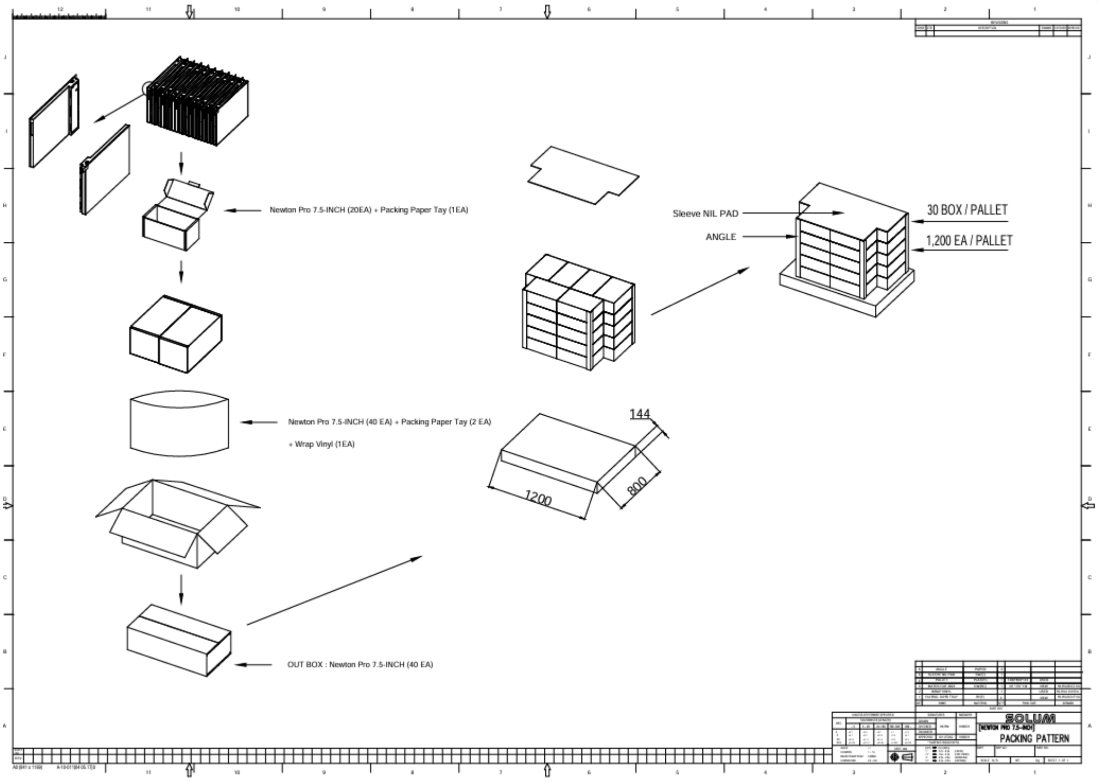


Figure 43. 7.5” Packing-Diagram (1,200 X 800)

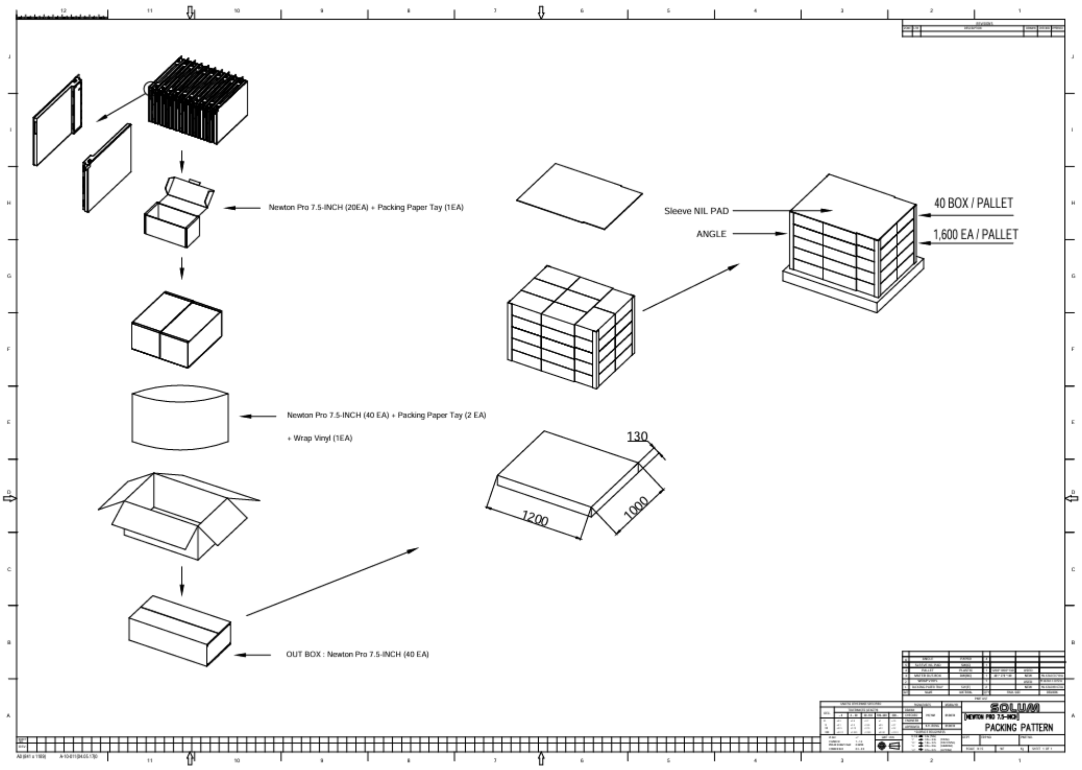


Figure 44. 7.5” Packing-Diagram (1,200 X 1,000)

4.13 9.7”

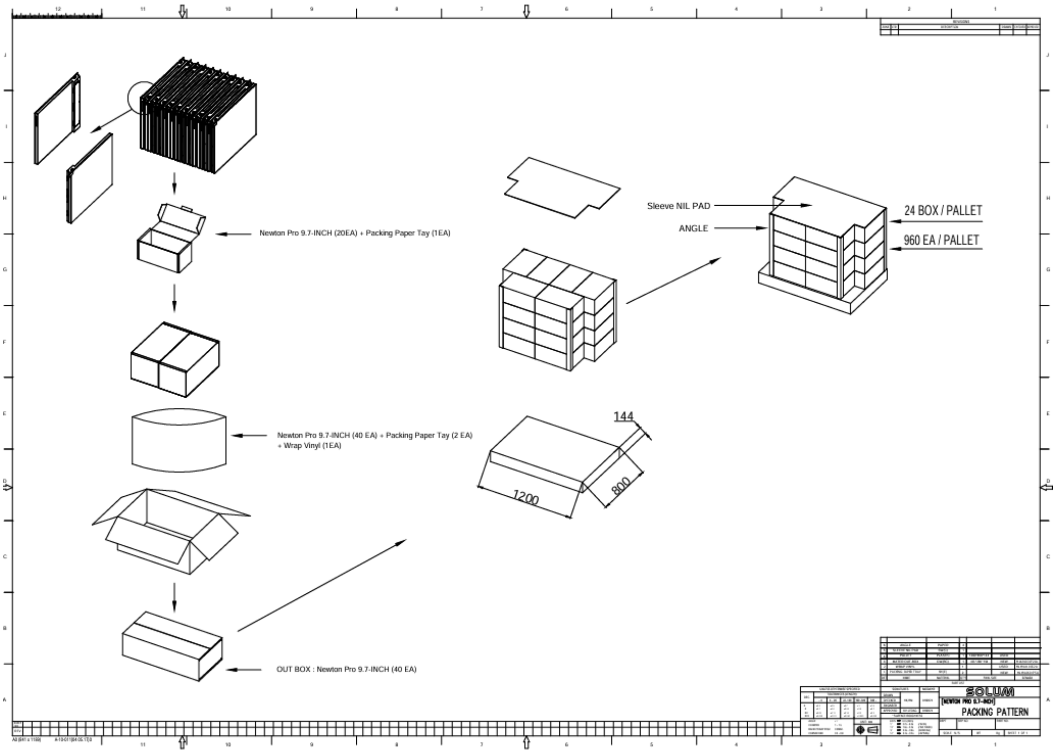


Figure 45. 9.7” Packing-Diagram (1,200 X 800)

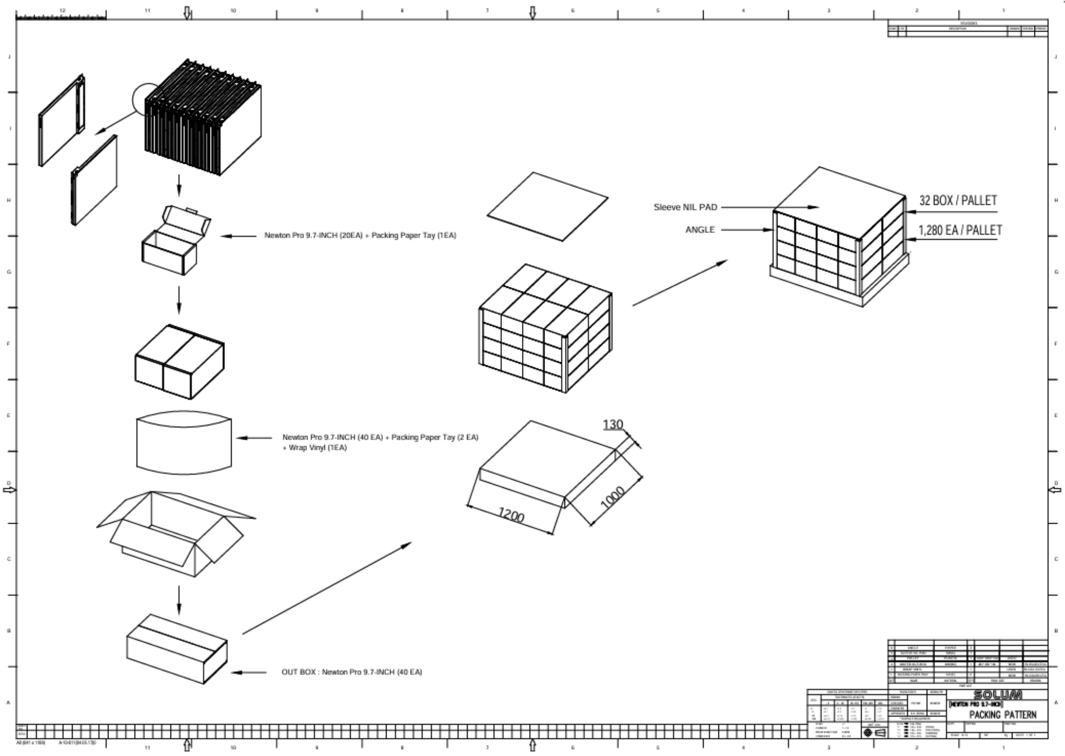


Figure 46. 9.7” Packing-Diagram (1,200 X 1,000)

4.14 11.6”

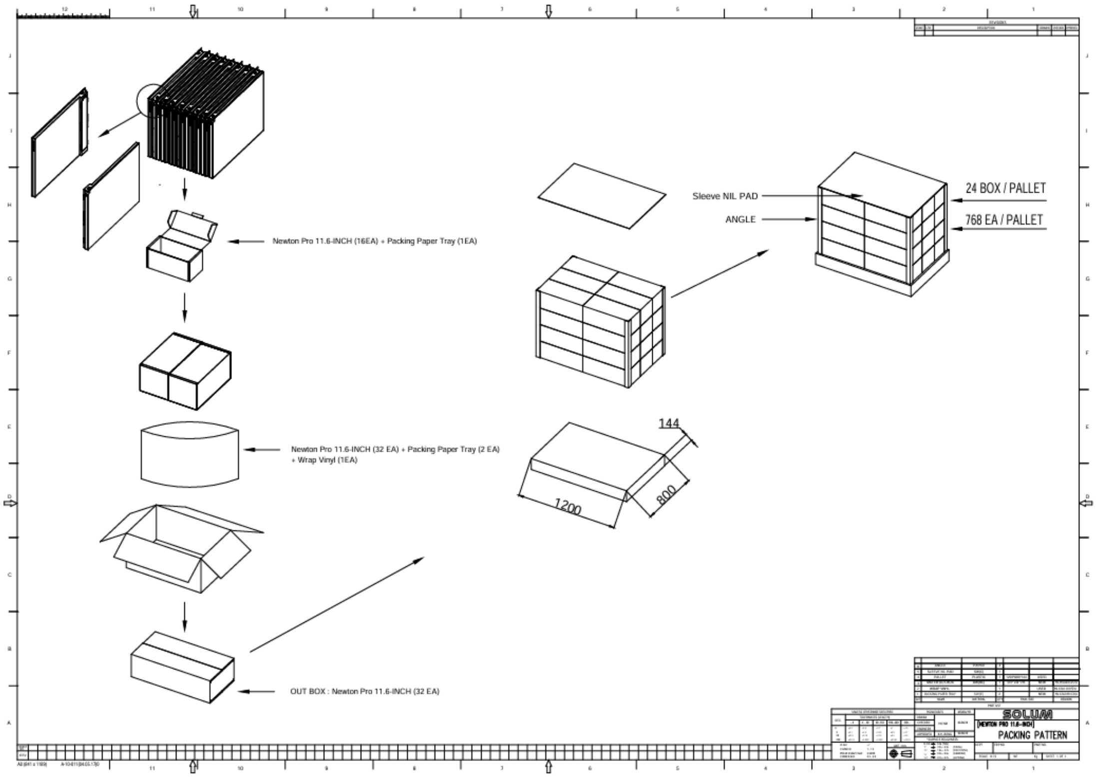


Figure 47. 11.6” Packing-Diagram (1,200 X 800)

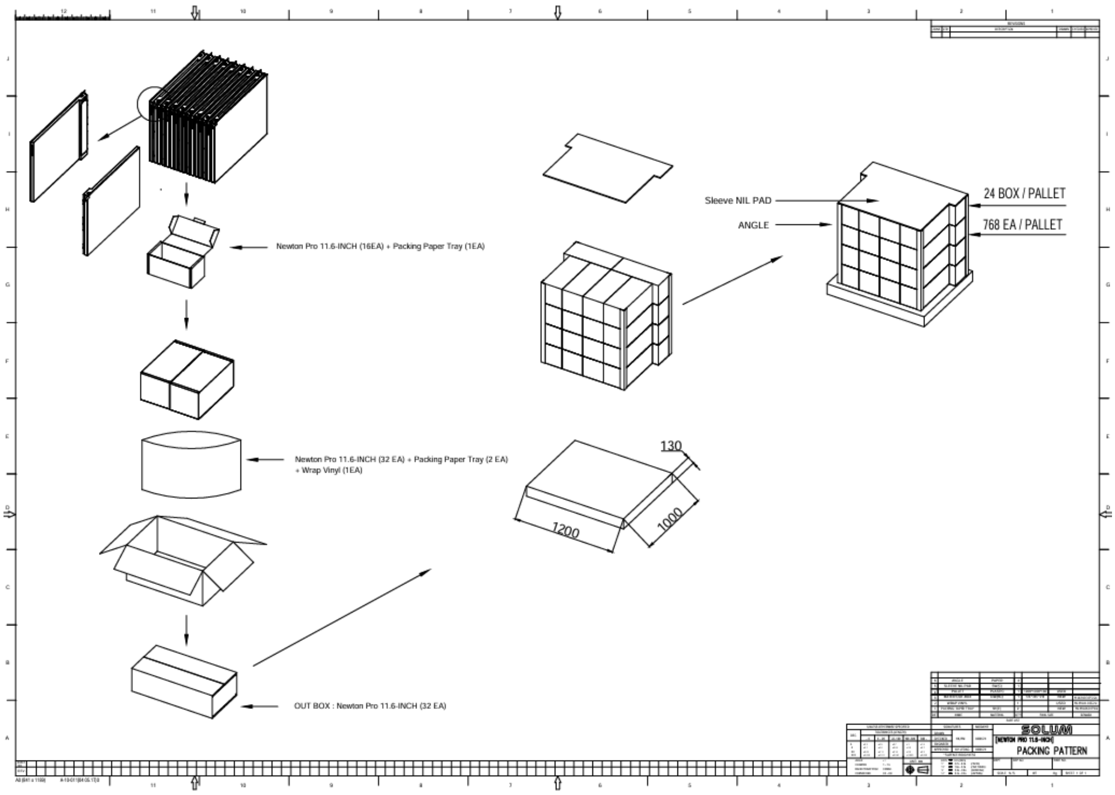


Figure 48. 11.6” Packing-Diagram (1,200 X 1,000)

4.15 12.2”

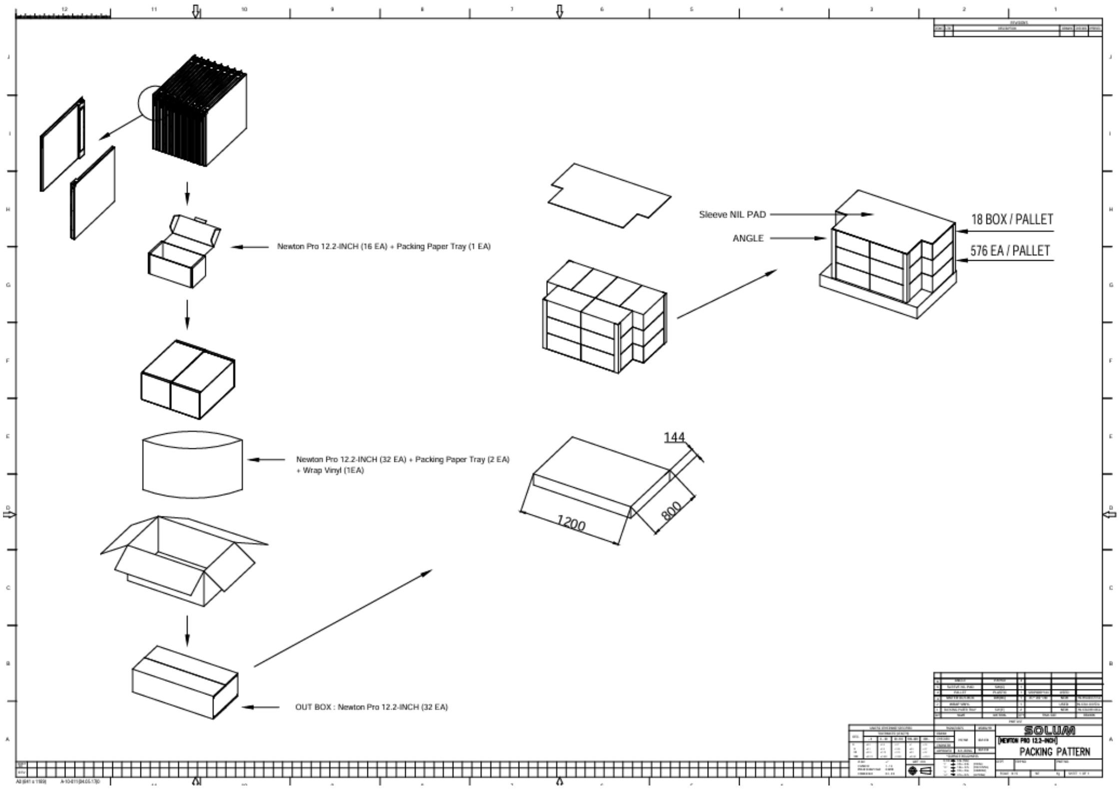


Figure 49. 12.2” Packing-Diagram (1,200 X 800)

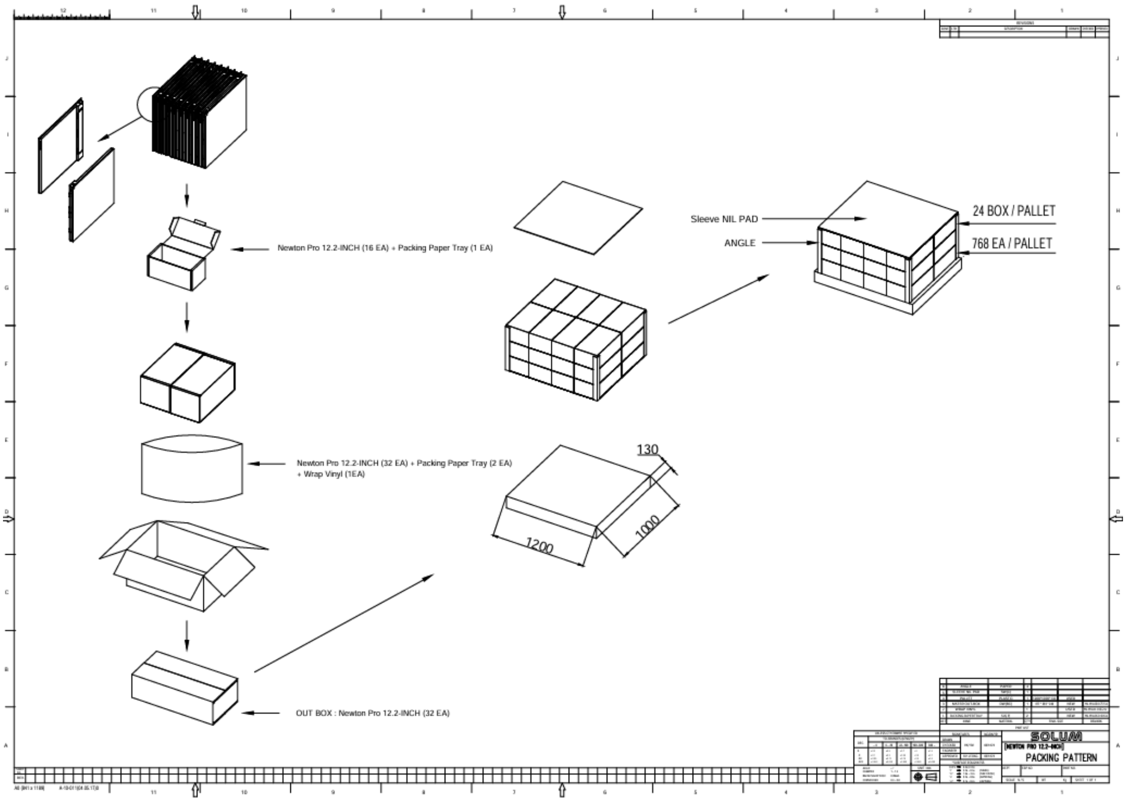


Figure 50. 12.2” Packing-Diagram (1,200 X 1,000)

5 Reliability Test

High Temperature Operation

Low Temperature Operation

High Temperature/Humidity Operation

High Temperature Storage

Temperature Shock (Storage)

ESD

Package Drop Test

Package Random Vibration Test

Test Item	Test Condition	Pass Criteria
High Temp Resistance	60℃ / 35%, 240hrs	Normal operation after test
High Temp Operation	40℃ / 35%, 240hrs	
Thermal Shock	-25℃ (for 30mins) ~ 60℃ (for 30mins) for 240 cycles	
High Temp & Humidity Operation	40℃ / 70%, 240hrs	
Low Temp Operation	Normal ESL: 0℃ (240hrs) Freezer ESL: -23℃ (240hrs)	
ESD Test	Air ±10KV, 150pF, 330Ω, 10 times/Point	
Waterproof (IEC60529)	Pre-treatment: Thermal shock(240hr), Drop (10 times), High Temp & High humidity (240hr), Button press (10 times) Waterproof Test: 15cm x 15cm water tank, 1m depth of water, 30 minutes (IP68 criteria) for NEWTON PRO tags only	No water ingress into tag
RF Sensitivity (Communication Distance)	Gateway <-> Tag distance: 100 meters	Tag receives RF signal from Gateway.

6 Product Handling Precautions

Provisions should be made to protect against any damage to the product caused by improper handling. The purchaser assumes any responsibility for damage to the product caused by improper handling.

Product should be stored in 32 °F ~ 104 °F (0°C ~ 40°C) @45~70% RH environment and should be installed within **90 days** of receipt.

6.1 Usage Environment

Take extra caution when using this RF device in the vicinity of other electronic devices and appliances. Most electronic devices and appliances use electromagnetic waves. Electromagnetic waves emitted by this RF device can affect other electronic devices and appliances.

If using the device in an explosion hazard area, follow all safety regulations, instructions, and signals.

6.2 Storage and Use

- The product is shipped in sleep mode (white screen), so it should be activated by pressing the button.
- Moisture and liquids can damage internal parts and circuit boards if allowed to enter into the device itself.
- Do not place or store the product on a sloped surface. The product may slide and fall off the surface and become damaged.
- Use the product in temperatures ranges of **0℃~40℃/32~104℉(BWRY), or -25℃~0℃/-13~32℉(Freezer)**. Parts and circuits may be damaged if operated or stored in extreme temperature.
- The display panel needs extra care during handling.
  - Do not apply any impacts on the e-Paper display as it is fragile.
  - Continuous exposure to excessive moisture (over 70% RH) or UV shortens display lifetime.
  - Ghosting image may appear in temperature conditions of less than **15℃/59℉ for normal tags and -25℃/ -13℉ for freezer tags**. (If ΔL* >2, we call it ghosting phenomenon)
- Avoid areas with strong magnetism or subject to magnetism.
  Contact between the device and a magnetic object can lead to malfunctions.
- Do not place the product near heat-producing kitchen appliances like a stove or a microwave or in the vicinity of highly pressurized containers.
- External impact to the product, such as from being dropped, can damage the product.

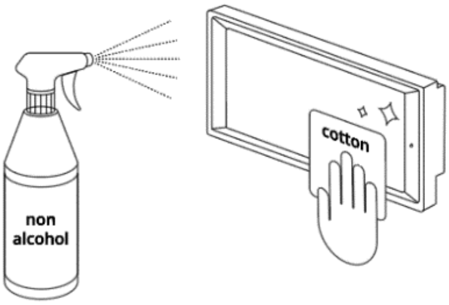
- Twisting and bending the product can damage the exterior casing and the internal components.
- If this product operates abnormally while removing battery or replacing battery, it needs to be discharged by contacting the battery terminals (+) and (-) in the product.
- This product uses the 2.4GHz frequency band for the wireless communication network. Radio communications can be limited or affected by other applications that share the same frequency band, such as WiFi, Bluetooth, Zigbee, etc.
- A prior investigation into the radio environment is strongly required for efficient and smooth installation.
- Frequent communications, updates and screen renewals may reduce battery life time.
- Low temperature environments may reduce battery life.
- FIFO (First In First Out)

6.3 Product Cleaning

For Spray Cleaning:

Steps

① Lightly spray all surfaces and wait a few seconds.
② Gently wipe clean using a cloth or tissue.
③ Let the labels dry.



Notes:

➢ Use mild, non-alcoholic detergents or glass cleaner.
➢ Recommend non-abrasive cloths: Microfiber, Cotton T-shirt, Cotton handkerchief, Cotton tea towel

For Wet Tissue Cleaning:

Steps

① Stand or lay down the labels.
② Wipe using wet tissues.
③ Let the labels dry.

6.4 Battery Replacement

Audience

- Authorized personnel with the following knowledge are allowed to replace the battery:
  Battery / Electronic assemblies (e.g. circuit board) / Compliance with the instruction

※ Note: Warranty is voided if battery is replaced by unauthorized personnel.
(When batteries require replacement, please contact the authorized personnel)

Instructions

- Risk of short circuit if battery is incorrectly installed/stored.
- Check that hands are dry before and at all times during the replacement process.
- Keep batteries away from children and infants.
- Do not heat, charge, bend, drop, short-circuit and/or disassemble battery.
- Do not mix together used and new batteries or different battery types.

※ Note: Battery rarely has minor stain or leak.

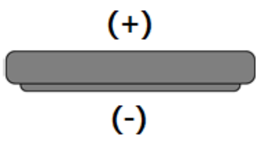
Steps

① Open the battery cover.
② Take out the batteries.
③ Put in the new batteries.
④ Check the batteries direction.
⑤ Put back in the battery cover.

Battery Direction

Top: (+) Positive
Bottom: (-) Negative

Battery (CR2450) ;



Red Wire: (+) Positive

Black Wire: (-) Negative

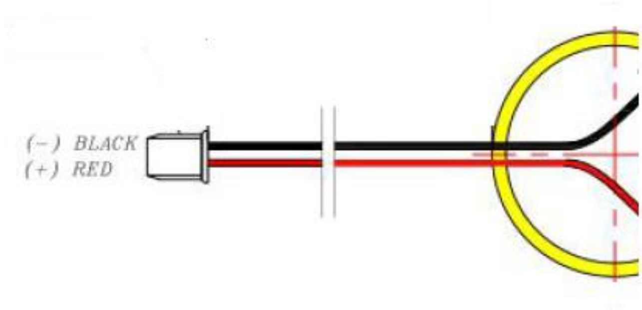


Figure 51. Battery Direction

7 Battery Handling Guide

7.1 Avoiding hazards in lithium battery handling

1. Do not short circuit (Fig. 1)

. Direct connection of plus (+) and minus (-) poles may result in leakage, heat generation, explosion and/or fire.

. Do not store and/or carry batteries with metallic items, such as a necklace.

2. Do not stack and/or jumble batteries (Fig. 2)

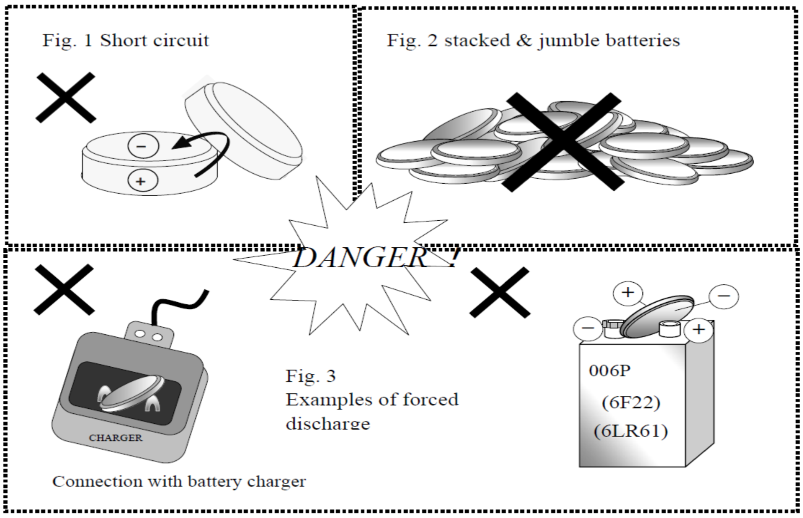
. Stacked and/or jumbled batteries may cause a short circuit and/or forced discharge from contact with other batteries.

. This may result in leakage, heat generation, explosion and/or fire.

3. Do not make forced discharge batteries (Fig. 3)

. On a forced discharge by an external power source, the battery voltage goes to negative and this causes gas generation in inside of the battery.

. This may result in leakage, heat generation, explosion and/or fire.



4. Do not dispose of batteries in fire

. Disposal of batteries in fire is extremely dangerous with a risk of explosion and violent flaring.

5. Do not heat batteries

. Heating batteries above 100℃/212°F may damage the resin in crimping, separator and other parts, potentially causing an electrolyte leak, internal short circuit, fire and/or explosion.

6. Do not solder directly onto batteries

. Direct soldering onto batteries may damage the resin in crimping, separator and other parts, potentially causing an electrolyte leak, internal short circuit, fire and/or explosion.

7. Do not recharge batteries

. Recharging of batteries may result in internal gas generation, causing electrolyte leak, battery swelling, fire and explosion.

8. Do not disassemble batteries

. Disassembly of batteries may generate gas that may irritate your throat.

. Lithium may also react with moisture to generate heat and fire.

9. Do not deform batteries

. Applying extreme pressure to batteries may cause deformation of the crimping and internal short circuit, causing electrolyte leak, battery swelling, fire and explosion.

10. Do not mix different type batteries

. For some applications, mixing different types of batteries or new and old batteries, can cause an overdischarge due to differences in voltage and discharge capacities.

. This may lead to the risk of swelling and/or explosion.

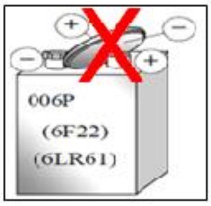
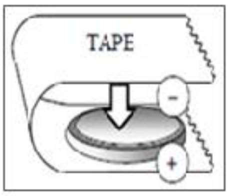
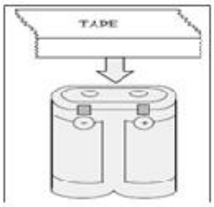
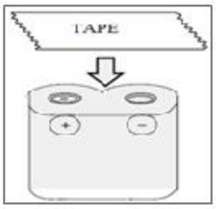
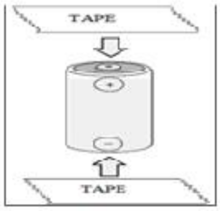
11. Do not insert batteries with opposite polarity

. For some applications, battery insertion with opposite polarity (reverse insertion of plus and minus) may result in leakage, heat generation, explosion and/or fire.

※ Please ensure the above precautions are strictly observed by related divisions including production, warehouse, product technology, sales, quality, customer stores, S/I companies, part-time workers, and external service companies.

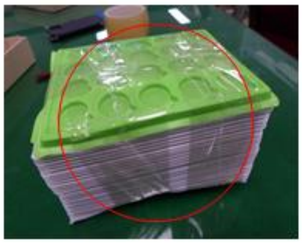
7.2 Proper Storing and Disposing of Lithium Batteries

■ To minimize risk of fire and explosion of batteries, be sure to follow the instructions below.

NG	Do not stack batteries	Do not stack batteries in a vinyl bags	Do not Short circuit	Do not make forced discharge batteries	
					
OK	Using battery tray → Tape sealing	Tape attachment to “+/- electrodes			
					

■ Proper use of battery tray is outlined below.

With batteries properly placed into each tray slot ➔ stack the trays in the same orientation ➔ use an empty tray on the top stack ➔ tape the stack together to avoid falling apart.

OK	Use the battery tray	Stack on the tray	Seal the tray with tape
			

■ Follow local regulations for proper battery disposal guideline.